

Monte Carlo Pinpoint: Stochastic Coordinate Localization via VLM Probing

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Abstract

For AI agents that autonomously operate graphical user interfaces, accurate coordinate localization on screen remains a fundamental challenge. The dominant approach, Set-of-Mark (SoM) prompting, relies on OpenCV contour detection and can only mark *UI elements*. However, real-world tasks often require localizing arbitrary semantic positions—the tip of a character’s nose, a specific object inside a painting, a defect on a textured surface—that are *not* UI elements at all.

We propose **Monte Carlo Pinpoint**, a three-phase method that converts coordinate estimation (a regression problem) into a ranking problem by asking a Vision–Language Model (VLM) “Which dot is closest to the target?” Phase 1 uses deterministic grid bisection to narrow the search to a coarse region ($\pm G$ pixels). Phase 1.5 elicits an *anchor declaration* from the VLM to stabilize subsequent judgments. Phase 2 places a deterministic seven-point hexagonal stencil inside the region, asks the VLM to select the closest dot with majority vote, and halves the search radius on each majority-HIT round (with bounded MISS retries that do not shrink the radius)—guaranteeing convergence when the target lies within the initial search disk and the VLM selects the closest dot on each HIT round.

We prove that the expected number of VLM queries is $O(v \cdot \log(s_0/\epsilon))$, where v is the majority vote size, s_0 the initial search radius, and ϵ the target precision—a logarithmic improvement over the $O(s_0^2)$ cost of exhaustive search. On a benchmark of 50 synthetic conditions across five categories, evaluated with a single VLM backbone and one trial per condition, Pinpoint—among the empirically evaluated Grid baseline and Pinpoint variants—is the only method in this evaluated set that achieves practical accuracy on the non-UI targets in this synthetic benchmark, while remaining competitive on standard UI elements; broader evaluation across additional VLM backbones, real-desktop tasks, repeat trials, and additional baselines (e.g., Direct/SoM-family methods) is reserved for future work (§9). We additionally distill a failure-mode taxonomy of VLM coordinate-localization hallucinations (the H-* family of seven cognitive failure classes) and benchmark-execution operational failures (the O-* family covering pipeline, rate-limit, credit-exhaustion, and audit-trail layers), with corresponding guardrail countermeasures grounded in real benchmark-execution incidents.

CCS Concepts

• **Human-centered computing** → HCI design and evaluation methods; • **Computing methodologies** → Artificial intelligence.

Table 1: Limitations of existing coordinate localization approaches.

Method	Principle	Limitation
Direct regression	Ask VLM for coords	Hallucination (± 50 –200 px)
SoM	OpenCV contours + labels	UI elements only
UI Automation	OS accessibility API	Programmable elements only
Template matching	Pre-registered images	Requires prior registration

Keywords

GUI agents, coordinate localization, vision–language models, Monte Carlo methods, visual prompting

1 Introduction

1.1 Problem Statement

GUI agents such as CogAgent, SeeClick, and Claude Computer Use possess the ability to “see the screen and take actions,” yet the *accuracy of coordinate localization* remains their principal bottleneck [5, 27]. When a VLM is asked directly for the coordinates of an on-screen element, it frequently *hallucinates*—producing estimates that are tens to hundreds of pixels away from the true location [31]. The very existence of dedicated Computer Use APIs is evidence that VLM-based coordinate regression cannot yet be trusted.

1.2 Limitations of Existing Approaches

Table 1 summarizes the landscape. The **core gap** is that no existing method can localize an *arbitrary semantic position inside an image* to pixel precision.

1.3 Contributions

This paper makes five contributions:

- (1) **Monte Carlo Pinpoint algorithm**: a three-layer architecture combining deterministic grid bisection, anchor declaration, and deterministic hex-stencil probing with HIT-only radius halving.
- (2) **Theoretical analysis**: derivation of the expected query count $O(v \cdot \log(s_0/\epsilon))$ via closest-dot convergence, including a model of VLM judgment error impact.
- (3) **Benchmark**: empirical evaluation across five categories (UI buttons, dense UI, natural images, ambiguous targets, precision tests) with 50 conditions.
- (4) **Open-source implementation**: released as the pinpoint tool in the video2ai-desktop MCP server (entry point: `mcp_server.py` at the repository root). The Phase 2 dot-placement primitive (`pinpoint_core.py`) is shared with the benchmark harness, preventing future dot-placement

drift between the two paths. Reported hex-stencil numerical results are reproduced from the 50-case rerun on the CLI subscription variant (Table 16).

- (5) **Failure mode analysis:** a taxonomy of VLM coordinate-localization cognitive failures (the H-* taxonomy of seven classes) and benchmark-execution operational failures (the O-* family covering pipeline, rate-limit, credit-exhaustion, and audit-trail layers), with matching pipeline-layer and audit-emission countermeasures grounded in a series of real benchmark-execution incidents and engineering experience.

2 Related Work

2.1 Direct Coordinate Regression

Several systems ask VLMs to output coordinates as text tokens. SeeClick [5] fine-tunes a VLM specifically for GUI grounding and proposes the ScreenSpot benchmark. UI-TARS [27] operates from screenshots alone with a coordinate-based action space. OS-ATLAS [38] compiles a 13 M-sample grounding dataset and finds 11.3% mislabeled entries in ScreenSpot. Recent work [31] demonstrates systematic spatial biases in VLM coordinate predictions, providing theoretical justification for Pinpoint’s regression-to-classification transform. Explicit position-to-coordinate mapping [33] analyzes the limits of generating coordinates as text tokens and improves regression through explicit position encoding; Pinpoint instead abandons regression entirely. Talking Points [28] pairs a point descriptor with a learned localizer for pixel-level grounding from natural-language descriptions; Pinpoint differs in remaining training-free and in deriving its precision from a Monte Carlo voting budget rather than from a regressed coordinate.

2.2 Fine-tuned Grounding Models

UGround [10] trains a universal GUI grounding model on large-scale synthetic web data. Aria-UI [42] achieves purely visual grounding without HTML or accessibility trees but depends on task-specific training. ShowUI [19] introduces UI-guided token selection for efficient screenshot processing. MolmoPoint [7] adds grounding-specific tokens to the VLM architecture. Pinpoint requires *no fine-tuning* and works with any off-the-shelf VLM.

2.3 Segmentation-Dependent Methods (SoM Family)

Set-of-Mark (SoM) [40] overlays numbered labels on regions detected by segmentation (SAM / OpenCV) and asks the VLM to select among them. It is the closest prior work to Pinpoint but is *inherently limited to segmentable regions*—i.e., UI elements. OmniParser [20] uses fine-tuned detection and captioning models, again limited to UI elements. MEGA-GUI [15] adopts a coarse ROI selection followed by fine localization, but relies on trained models. Pinpoint replaces segmentation with a deterministic hex-stencil dot arrangement, eliminating the dependency on element detectors.

2.4 Zoom and Iterative Refinement

This family of methods is conceptually closest to Pinpoint’s Phase 1. RegionFocus [21] zooms into candidate regions at test time and re-queries the VLM—training-free but lacking a probabilistic refinement phase. AdaZoom-GUI [26] adapts zoom level to element size and refines instructions at each step, resembling Phase 1 + Phase 1.5 but without Monte Carlo refinement. A systematic study of zooming [12] confirms that zoom improves GUI grounding accuracy but does not address how to go beyond the zoom limit. GUI-Spotlight [16] uses iterative spotlighting with think-with-image, requiring training. Chain-of-Ground [17] narrows the search space through iterative reasoning steps in text; Pinpoint uses *visual* overlays (grids, dots) as the reasoning medium. Several contemporaneous works extend the zoom paradigm with explicit uncertainty signals: UI-Zoomer [30] drives zoom selection by token-distribution uncertainty, AutoFocus [43] uses token-level perplexity to model spatial uncertainty for active visual search, Zoom Consistency [14] shows that geometric agreement across zoom steps is itself a calibration-free confidence signal, and BAMI [45] addresses precision and ambiguity biases in GUI grounding through training-free coarse-to-fine focusing with candidate selection; Pinpoint instead uses deterministic grid bisection followed by Monte Carlo voting, deriving its certainty guarantee from a sample budget rather than from a learned uncertainty estimator or a focusing heuristic.

2.5 Test-Time Voting and Scaling

GUI-RC [8] constructs a spatial voting grid from multiple predictions and uses region consistency—the most directly related work. However, GUI-RC aggregates predictions *across queries*, whereas Pinpoint converts the problem to classification *within a single query*. DiMo-GUI [36] provides a training-free framework with dynamic visual grounding and modality switching. GTA1 [41] learns test-time scaling via RL; Pinpoint is purely inference-time. GAIA [32] uses best-of-*N* selection, conceptually similar to Pinpoint’s majority vote but over complete action predictions rather than dot classifications.

2.6 Coordinate-Free Approaches

GUI-Actor [37] replaces coordinate regression with an attention-based action head. While it shares Pinpoint’s problem recognition (coordinate regression is ill-suited for VLMs), it requires *architectural changes*; Pinpoint is a pure prompting strategy applicable to any VLM.

2.7 Visual Prompting

VP-VLA [34] overlays spatial anchors (crosshairs, bounding boxes) on visual input for robotics tasks, sharing Pinpoint’s philosophy of augmenting images with visual prompts. V2P [4] uses background suppression and center peaking to improve coordinate regression—still a regression approach addressed via data augmentation, whereas Pinpoint replaces regression with classification.

2.8 Agent Orchestration and Harnesses

Concurrent with this work, a wave of open-source orchestration specs and harness patterns has emerged at the systems level: OpenAI’s Symphony [24, 25], released 2026-04-27 under Apache 2.0,

treats a Linear project board as the control plane for autonomous Codex agents, with workflow loaders, retry backoff, and structured logging surfaces; alongside this, industry harnesses such as Anthropic’s three-agent pattern [11] decompose long-running tasks across planner / generator / evaluator phases (analysed further in Section 8.2). Symphony’s scope is full task lifecycle orchestration—issue pickup, agent restart, PR landing—operating at the workflow-orchestration layer, above the per-task agent layer that hosts primitives like Pinpoint. Pinpoint, by contrast, uses agentic tool-use as a *probing primitive*: a single coordinate query is wrapped in a structured tool call (Section 5.5, the CLI subscription variant), and $N=3$ plurality voting on tool-use responses (the CLI plurality-vote variant) yields measurable selectivity gains (Section 8.3). The two layers compose: a Symphony-style orchestrator could dispatch Pinpoint as a coordinate-finding tool. We position Pinpoint as the lower-level visual-grounding component such an orchestration stack would invoke.

Concurrent work by Augment Code introduces “Intent” [2], an agentic development environment that “coordinates agents through a shared living spec rather than wiring them together through code”. Intent employs a Coordinator–Implementor–Verifier topology—a Coordinator agent proposes a plan as a spec, Implementor agents execute in parallel waves, and a Verifier checks results against the spec—and thus shares the three-role harness structure under which Pinpoint was developed. Two axes distinguish the present work from such spec-driven orchestrators: a heterogeneous-model consultant tier that supplies independent cross-validation of same-model reviewer agreement (Codex CLI in our deployment), and a visual-plus-text dual-surface programming model that neither pure-CLI orchestrators (e.g., Symphony or Intent) nor pure-visual workflow platforms occupy.

2.9 System-layer Computer Use and GUI Grounding

Beyond the algorithmic sub-pixel grounding primitives we survey above, a complementary research line addresses the *system layer*: how an agent should perceive, navigate, and click within full desktop and browser environments. OpenAI’s Computer-Using Agent [23] reports OSWorld 38.1 %, WebArena 58.1 %, and WebVoyager 87.0 % (self-reported, January 2025). Anthropic’s computer-use tool [1] provides 1 : 1 native coordinate handling for screenshots up to 2576 px on the long edge with an explicit zoom action for inspecting sub-regions at full resolution. On the high-resolution professional grounding benchmark ScreenSpot-Pro [18], accuracy has progressed from an initial 18.9 % to ScreenSeeker’s 48.1 % via cascaded visual search, GUI-Actor-7B + Qwen2.5-VL-7B’s 44.6 % via coordinate-free attention [37], OSWorld-G/Jedi-7B’s 39.5 % via grounding-data scaling [39], and RegionFocus’s 61.6 % via test-time zoom on a Qwen2.5-VL-72B base [21]. The published values remain below human level for the highest-resolution targets, and apples-to-apples comparison is complicated by base-model size and search-augmentation variation. Pinpoint, the algorithm-layer primitive we present, is positioned to plug into any of these system-layer harnesses as a deterministic coordinate-inference fallback for boundary-ambiguous and sub-pixel targets that remain difficult for end-to-end VLMs.

Table 2: Comparison of approaches. Pinpoint combines deterministic grid narrowing, anchor declaration, and hex-stencil dot voting—a combination absent from all prior work.

Approach	Representative	Pinpoint Advantage
Direct regression	SeeClick, UI-TARS	Avoids hallucination via classification
Fine-tuning	UGround, Aria-UI	Training-free; any VLM
Segmentation-dep.	SoM, OmniParser	No detector; non-UI targets
Zoom refinement	RegionFocus	Probabilistic phase beyond zoom limit
Test-time voting	GUI-RC, GAIA	Intra-query classification voting
Architecture change	GUI-Actor	Prompting only; no model changes

2.10 Agent Failure Taxonomies and Incident Analysis

Beyond the algorithm-layer and system-layer prior art surveyed above, a parallel literature classifies the failures of *agent systems themselves*. We position our H-*/O-* split against five lines of this work. Recent multi-agent-system work classifies traces into system-design, inter-agent-misalignment, and task-verification failures [3]; our taxonomy instead separates VLM-internal visual grounding hallucinations from operational failures in the surrounding benchmark and agent infrastructure. At the single-agent level, AgentErrorTaxonomy localizes failures across memory, reflection, planning, action, and system-level modules [46], whereas our H/O split isolates visual-grounding hallucinations from non-model operational faults in the benchmark harness. On the software-engineering side, empirical studies of agentic-AI repositories classify faults by type, symptom, root cause, and architectural dimension [29]; our O-* family narrows that lens to the operational layers that affected a VLM grounding benchmark. Agent-incident-analysis frameworks distinguish system-related, contextual, and cognitive factors [9]; our taxonomy operationalizes a related split at benchmark scale by separating VLM coordinate-localization hallucinations from pipeline, rate-limit, credit-exhaustion, and audit-trail failures. Two transferred-discipline analogues motivate the layered diagnostic stance: HFACS [35] separates immediate unsafe acts from latent organizational conditions, a stance our O-* stack applies to benchmark-execution workflows; and ODC [6] treats defect classifications as in-process measurement signals, mirroring how separating H-* and O-* preserves whether accuracy loss came from model grounding or operational infrastructure. None of the above provides the specific cognitive-vs-operational separation we propose for VLM grounding systems, but each informs where our taxonomy sits in the broader agent-failure analysis landscape.

2.11 Positioning of This Work

Table 2 summarizes the positioning. No existing method combines all three phases of Pinpoint: deterministic grid narrowing, anchor-based self-consistency, and hex-stencil dot voting. The use of distinctly colored hex-stencil dots for *intra-query regression-to-classification* conversion is novel.

3 Method

3.1 Problem Formalization

Input: An image I and a natural-language target description T (e.g., “the center of the gear icon”).

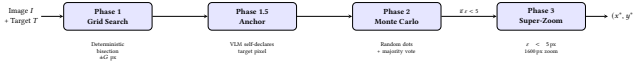


Figure 1: Overview of the Monte Carlo Pinpoint pipeline. Phase 1 uses deterministic grid bisection; Phase 1.5 elicits an anchor declaration; Phase 2 applies deterministic hex-stencil probing with HIT-only radius halving and majority vote; Phase 3 optionally refines to 1 px precision.

Output: Target coordinates $(x^*, y^*) \pm \varepsilon$ pixels.

Oracle: A VLM $V(I', Q) \rightarrow A$, which returns an answer A to a textual question Q given a (possibly annotated) image I' .

Core transformation: We convert coordinate estimation (regression) into the ranking question “Which dot is closest to the target?”

3.2 Algorithm Overview

The algorithm proceeds in three (optionally four) phases:

Phase 1 – Deterministic Grid Search. Narrows the target to a $\pm G$ -pixel region via axis-independent binary search over grid cells.

Phase 1.5 – Anchor Declaration (Guardrail). Elicits a precise natural-language self-declaration from the VLM of *exactly which pixel* it aims at, anchoring subsequent judgments.

Phase 2 – Hex-Stencil Dot Probing. Places a deterministic seven-point hexagonal stencil ($k=7$) in the region, asks the VLM which dot is closest to the anchor, and halves the search radius on each majority-HIT round (with bounded MISS retries that do not shrink the radius); converges under the containment and judgment assumptions stated in Section 4.4.

Phase 3 – Super-Zoom Refinement (Optional). For $\varepsilon < 5$, re-probes at extreme magnification around the Phase 2 hit point to reach 1 px accuracy.

Figure 1 illustrates the flow.

3.3 Phase 1: Deterministic Grid Search

The X and Y axes are searched independently via binary bisection. At each step, n labeled grid lines are drawn on the image and the VLM is asked “Which line is closest to the target?” The VLM replies with either an exact line (“L5”) or a between-lines interval (“B3-4”). Each response narrows the search region by a factor of $\sim n$.

Two refinement mechanisms improve accuracy:

- **Force zoom on first exact hit.** When the VLM reports an exact line on the first iteration, the result is treated as a between-lines interval spanning the adjacent lines (e.g., L5 $\rightarrow$ B4-6). This forces a zoom refinement pass, mitigating VLM overconfidence at coarse scale.
- **Zoom mode.** When the search interval drops below $w/3$ pixels, the region is cropped and magnified to 1200 px before re-querying. This provides the VLM with higher-resolution input for the final bisection steps. Image dimensions are clamped to 7900 px to respect API limits.

Failure fallback. If the grid search fails to locate the target on either axis (VLM returns “NONE”), the algorithm falls back to a single direct VLM coordinate estimation query for the failed axis. This adds one query but prevents catastrophic failure where the search center defaults to the image midpoint.

Each axis requires 2–4 VLM queries (typically 2 normal + 1–2 zoom). Temperature is set to 0 for reproducibility.

3.4 Phase 1.5: Anchor Declaration

Before entering the stochastic phase, we stabilize the VLM’s judgment criterion:

- (1) The Phase 1 region is cropped and magnified.
- (2) The VLM is asked to *declare* in natural language the exact pixel it aims at—e.g., “the gear icon” becomes “the exact center pixel of the gear icon’s hub circle.”
- (3) This declaration is used as the judgment criterion for all Phase 2 rounds.
- (4) On the ABSENT wider-crop retry path, the anchor is re-requested; periodic re-declaration and semantic drift comparison are reserved as future / interactive-server guardrails.

Failure recovery. If the VLM cannot identify the target in the initial crop (detected by responses beginning with “I cannot locate...”), the crop is widened to $\pm w/3$ pixels (one-third of the image dimension) and the anchor is re-requested. This handles cases where Phase 1’s grid estimate is sufficiently inaccurate that the target falls outside the narrow context crop.

Rationale. VLMs tend to *drift* in their interpretation of the target across multiple rounds. An explicit anchor declaration forces consistency and enables metacognitive correction when drift is detected.

3.5 Phase 2: Monte Carlo Dot Probing

Phase 2 uses a *closest-dot convergence* strategy with a deterministic seven-point hexagonal half-net rather than a binary HIT/MISS test. Each round proceeds as follows:

- (1) Place the seven-point hexagonal stencil (one center dot, six on the ring of radius $\frac{\sqrt{3}}{2}s$ at 60° increments) inside the current search disk centered at $(\hat{x}, \hat{y})$ with radius s .
- (2) Crop the context area ($3 \times s$) and magnify to a no-loss capped image (for Claude-class VLMs, 1024 px on the longer axis).
- (3) Annotate each dot with a numbered label and a unique color (adaptively selected for background contrast).
- (4) Ask the VLM: “Which dot is closest to the anchor position?” The VLM *must* pick one—there is always a closest dot, and by Theorem 4.1 the chosen dot lies within $s/2$ of the target under closest-dot semantics.
- (5) **Majority vote:** repeat the query $v = 3$ times; accept only if ≥ 2 agree on the same dot.
- (6) On consensus (HIT), recenter on the chosen dot and halve the search radius: $s \leftarrow \max(2, \lfloor s/2 \rfloor)$. On no consensus (MISS), retry with the same (center, s) up to 3 consecutive times; if all 3 retries also MISS, Phase 2 halts and the algorithm returns the Phase 1 estimate.

The hex stencil’s $s/2$ -cover guarantee (Fig. 2) ensures the chosen dot on a HIT is close enough that recentering and halving preserves containment. After n HIT rounds the search radius follows the integer-clamped recurrence $s_n = \max(2, \lfloor s_{n-1}/2 \rfloor)$, converging to the 2-pixel floor in $\lceil \log_2(s_0/2) \rceil$ rounds.

Design choices. Dot radius is 4 px (on the magnified image) for visibility, with number labels offset to avoid overlap. Dot colors are *adaptively selected* based on background brightness: bright colors for dark backgrounds, saturated dark colors for light backgrounds, and default high-saturation colors otherwise. A minimum spacing constraint ensures dots are ≥ 2 px apart from each other. The zoom size of 1200 px provides sufficient resolution for the VLM to distinguish fine details.

3.6 Phase 3: Super-Zoom Refinement

When the requested precision $\varepsilon < 5$ px, Phase 3 takes the Phase 2 hit point and magnifies a $\pm(\varepsilon + 2)$ px region to 1600×1600 px. Dot radius is reduced to 1, and the probing is repeated to achieve the theoretical limit of 1-pixel accuracy.

4 Theoretical Analysis

4.1 Expected Query Count

Phase 2 uses a closest-dot convergence strategy with a deterministic seven-point hexagonal half-net. Each round, the algorithm places seven dots in the current search disk: one at the center and six on a ring of radius $\frac{\sqrt{3}}{2}s$ at angles $0^\circ, 60^\circ, \dots, 300^\circ$ (Fig. 2). The VLM is asked to pick the dot closest to the target. On a majority HIT the search center moves to the chosen dot and the search radius halves; on a MISS (no consensus) both the center and the radius are preserved and the round is retried with the same stencil. Halving therefore occurs on HIT rounds only.

THEOREM 4.1 (CONVERGENCE OF PHASE 2 UNDER HEXAGONAL HALF-NET). *Let D_n denote the search disk on HIT round n , with radius s_n . The seven-point hexagonal stencil placed in D_n is an $\frac{s_n}{2}$ -cover of D_n : every point in D_n lies within $\frac{s_n}{2}$ of at least one stencil dot.*

Under closest-dot semantics combined with v -vote majority and HIT-only halving, the algorithm tolerates up to 3 consecutive MISS retries per HIT round and halts on the 4th consecutive MISS, returning the Phase 1 estimate. Provided the target is contained in D_0 , the returned center after n HIT rounds is within $s_n = s_0/2^n$ of the target. Each MISS round retries with the same (center, s) and does not advance the search radius. The number of HIT rounds needed to reach precision ε from initial radius s_0 is

$$n_{\text{HIT}} = \left\lceil \log_2 \left(\frac{s_0}{\varepsilon} \right) \right\rceil. \quad (1)$$

Proof sketch. (i) The cover claim is geometric: the worst-case point in a unit disk relative to the stencil is the disk boundary at the angular midpoint between two outer dots, e.g. $(\frac{\sqrt{3}}{2}, \frac{1}{2})$, whose distance to the nearer outer dot $(\frac{\sqrt{3}}{2}, 0)$ is exactly $\frac{1}{2}$; interior points within radius $\frac{1}{2}$ of the center are covered by the center dot. Scaling by s_n gives the $\frac{s_n}{2}$ -cover. (ii) On a HIT, the cover places the chosen dot within $s_n/2$ of the target; recentering on the dot and halving the radius keeps the target inside D_{n+1} . This is the only step that advances n . (iii) On a MISS, the algorithm retries with the same (center, s_n); containment in D_n is unchanged, the retry counter

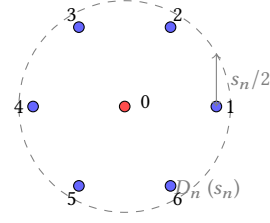


Figure 2: Phase 2 hexagonal half-net (seven dots). One dot at the disk center plus six on the ring of radius $\frac{\sqrt{3}}{2}s_n$ at 60° increments. The maximum distance from any point in D_n to the nearest dot is $\frac{s_n}{2}$, achieved at the angular midpoint between adjacent outer dots. This is the $\frac{s_n}{2}$ -cover guarantee that Theorem 4.1 relies on.

increments, and Phase 2 halts on the 4th consecutive MISS (i.e. after up to 3 tolerated retries within the current HIT round), returning the Phase 1 estimate. (iv) Termination: the outer loop is bounded by HIT count $n_{\text{HIT}} \leq \max_rounds$ and the inner retry budget is bounded by 3, so the total query batches are $O(n_{\text{HIT}})$ in the typical case and $O(1)$ on early-MISS exhaustion. By induction on n , the returned center after n HIT rounds is within $s_0/2^n$ of the target, as claimed. $\square$

The convergence rate in Theorem 4.1 is *deterministic in HIT count*: the halving occurs on HIT rounds only and the $\frac{s_n}{2}$ -cover guarantee fixes the per-HIT shrink factor. The VLM’s accuracy affects how many query batches Phase 2 spends per HIT (consecutive MISSes consume retry budget without advancing the radius), not the per-HIT shrink rate.

Phase 2 issues query batches of v VLM votes each (majority vote). Let n_{HIT} denote the number of *accepted* HIT rounds (the round counter advanced by Theorem 4.1) and n_{MISS} the number of MISS retry batches issued; Phase 2’s total batch count is $B_{\text{Phase2}} = n_{\text{HIT}} + n_{\text{MISS}}$. In the empirical-result tables, n_{HIT} is reported as `phase2_hit_rounds` and B_{Phase2} as `phase2_rounds`. The associated query count is

$$Q_{\text{Phase2}} = B_{\text{Phase2}} \times v. \quad (2)$$

The Phase 2 batch budget depends on the trajectory class:

- **No-MISS lower bound** (clean trajectory): every round is a HIT, so $B_{\text{Phase2}} = n_{\text{HIT}} = \lceil \log_2(s_0/\varepsilon) \rceil$.
- **Successful non-terminal trajectory** (every HIT round accepts within at most 3 MISS retries): $B_{\text{Phase2}} \leq 4 n_{\text{HIT}}$.
- **Terminal MISS exhaustion** (Phase 2 halts on the 4th consecutive MISS within an unaccepted round): the final unaccepted round contributes up to 4 batches with no HIT advance, so $B_{\text{Phase2}} \leq 4 n_{\text{HIT}} + 4$. This handles edge cases such as MMMM ($B = 4, n_{\text{HIT}} = 0$) and HMMMM ($B = 5, n_{\text{HIT}} = 1$), which violate the unqualified $4 n_{\text{HIT}}$ bound.
- **Absolute implementation bound** (independent of trajectory class): $B_{\text{Phase2}} \leq 4 \max_rounds = 32$ at the default schedule, so $Q_{\text{Phase2}} \leq 96$ at $v = 3$.

Table 3: Default parameter settings and resulting query counts.

Parameter	Symbol	Value
Screen width	N	1920 px
Grid precision	G	20 px
Initial search radius	s_0	100 px
Dots per round	k	7
Majority vote size	v	3
Max rounds		8
Phase 1 queries (2 axes)		$\approx 4-8$
Anchor query		1
Phase 2 queries, no-MISS (8×3)		24
Phase 2 queries, worst-case ($\leq 32 \times 3$)		≤ 96
Total, no-MISS		$\approx 29-33$
Total, worst-case		$\leq 101-105$

Table 4: Scaling comparison: exhaustive search vs. Monte Carlo Pinpoint Phase 2 ($v = 3$ votes per round).

Method	$s_0=20$	$s_0=100$	Complexity
Exhaustive (worst)	400	10,000	$O(s_0^2)$
Pinpoint Phase 2	~ 15	~ 21	$O(v \cdot \log(s_0/\epsilon))$
Speedup	$27\times$	$476\times$	$O(s_0^2/\log s_0)$

Combining Phase 1, the anchor declaration, and Phase 2, the total query count for a full Pinpoint run is

$$Q_{\text{pinpoint}} = \underbrace{2 \times (2-4)}_{\text{Phase 1}} + \underbrace{1}_{\text{Anchor}} + \underbrace{B_{\text{Phase2}} \times v}_{\text{Phase 2}}, \quad (3)$$

with B_{Phase2} bounded by the trajectory-class cases listed above; in particular the absolute implementation bound is $B_{\text{Phase2}} \leq 32$ for the default schedule ($\text{max_rounds} = 8$), where s_0 is the initial Phase 2 search radius, set to $s_0 = 100$ px in the evaluated implementation to absorb the ~ 68 th-percentile Phase 1 error envelope. Earlier drafts of this paper used $s_0 = G + 2$ (tying the Phase 2 radius to the Phase 1 grid pitch); the widened constant decouples Phase 2 convergence from grid-pitch drift.

4.2 Numerical Example

Table 3 shows that the default configuration’s no-MISS lower bound is approximately 29 VLM queries to localize a target on a 1920-pixel-wide screen, with an absolute implementation upper bound of ≤ 105 queries when MISS retries are exhausted on every HIT round. The measured mean of 29.1 queries closely matches the no-MISS lower bound, indicating MISS retries are rare on the evaluation set.

4.3 Impact of VLM Judgment Error Rate

Let p denote the probability that the VLM correctly identifies the closest dot to the target. An incorrect selection causes the search center to move toward a non-target dot, degrading convergence direction while the radius still halves.

Without majority vote ($v = 1$): each round has probability $(1 - p)$ of moving toward a wrong dot. After n rounds, the expected number of correct moves is np , giving a random-walk component that increases final error.

With majority vote ($v = 3$, majority rule, subject to the H-2 pseudo-majority caveat at Section 4.4): the probability of a correct consensus is

$$P_{\text{correct}} = 3p^2(1 - p) + p^3 = p^2(3 - 2p). \quad (4)$$

For $p = 0.8$, $P_{\text{correct}} = 0.896$ (a ~ 10 percentage-point improvement). For $p = 0.9$, $P_{\text{correct}} = 0.972$. When no consensus is reached (all three votes disagree), the center and search radius are unchanged and the round is retried with the same hex stencil. A per-HIT-round retry counter caps consecutive MISSes at 3; on the 4th MISS in a row Phase 2 halts and the algorithm returns the Phase 1 estimate. Halving is HIT-only: the inductive containment invariant in Theorem 4.1 requires that the search radius shrink only after a chosen dot has been placed within $s/2$ of the target.

Practical implication. With $v = 3$ and 8 max rounds, the algorithm tolerates up to ~ 2 incorrect rounds while still converging within 20 px of the target from $s_0 = 100$ px (the continuous-ideal $s_8 = 100/256 \approx 0.39$ px upper bound is a useful intuition; under the integer-clamped recurrence of equation (5) the radius reaches the 2 px floor in 6 HIT rounds and the actual final-error bound is given by Theorem 4.2).

4.4 Convergence Guarantee for Phase 2

The hex-stencil closest-dot strategy guarantees convergence in search radius across HIT rounds, conditional on (i) the target lying within the initial Phase 1 search disk and (ii) the VLM correctly selecting the closest dot on each HIT round; §6.3 (H-2 Pseudo-majority vote) documents the empirical caveat to (ii) under temperature-zero deterministic decoding, where Table 5’s independent-trial assumption is violated. In implementation, the radius follows the integer-clamped recurrence

$$s_{t+1} = \max(2, \lfloor s_t/2 \rfloor), \quad (5)$$

so for $s_0 = 100$ px the actual sequence is $100 \rightarrow 50 \rightarrow 25 \rightarrow 12 \rightarrow 6 \rightarrow 3 \rightarrow 2 \rightarrow 2 \rightarrow 2$, reaching the floor of 2 px after 6 HIT rounds. The continuous bound $s_n = s_0/2^n$ is tight up to a 1-pixel rounding slack and is the right asymptotic.

THEOREM 4.2 (ACCURACY BOUND UNDER CORRECT VLM JUDGMENTS). *If the VLM correctly selects the closest dot on every HIT round, the final error after n HIT rounds is bounded by the hex-stencil cover at the radius used to place the n -th HIT (i.e. the pre-halving radius s_{n-1}):*

$$d_{\text{final}} \leq \frac{s_{n-1}}{2} + \delta_{\text{grid}}, \quad (6)$$

where $\delta_{\text{grid}} \in [0, 1)$ px absorbs both the floor-rounding slack of equation (5) and the integer-coordinate clipping applied to stencil dots near image edges. For $s_0 = 100$, $n = 6$: the 6th HIT is selected from the stencil built at $s_5 = 3$ px, so $d_{\text{final}} \leq 1.5$ px + δ_{grid} . After the 6th HIT the stored radius drops to the 2 px floor, so subsequent HITs (rounds 7 and 8 in the default schedule) cannot sharpen the bound below ≈ 1 px without changing the floor.

In practice, VLM judgment errors prevent reaching this theoretical limit. Table 5 shows the degradation.

Table 5: Final error as a function of VLM accuracy p ($s_0 = 100$, $n = 8$, $k = 7$, $v = 3$). With majority vote, even moderate VLM accuracy yields practical precision.

p	$P_{\text{correct}} (v=3)$	Expected wrong rounds	Typical error
1.0	1.000	0.0	<1 px
0.9	0.972	0.2	~5 px
0.8	0.896	0.8	~15 px
0.7	0.784	1.7	~40 px

Table 6: Benchmark categories.

Category	Description	SoM Capability
ui_button	Standard UI buttons/icons	Strong
ui_dense	Dense UI (toolbars, tab bars)	Moderate
natural	Non-UI (semantic points in natural images)	Impossible
ambiguous	Low-contrast / small / vague targets	Weak
precision	Small/fine targets (2–60 px features)	Weak

5 Experimental Evaluation

5.1 Benchmark Design

We construct a benchmark of **5 categories** $\times$ **10 conditions each** = **50 test cases**, generated synthetically (generate_test_images.py) with known ground-truth pixel coordinates.

5.2 Compared Methods

- (1) **Direct** (discussed for context; not evaluated empirically in this work): Ask the VLM “What are the coordinates of X?” The hallucination failure mode is structural (Table 1); we omit an empirical row because the comparison would not contribute new information.
- (2) **SoM** (discussed for context; not evaluated empirically in this work): OpenCV contour detection $\rightarrow$ numbered labels $\rightarrow$ VLM selection [40]. SoM is constrained to UI elements detectable by contour analysis (Section 2).
- (3) **Grid** (empirically evaluated): Deterministic grid bisection only (equivalent to Pinpoint Phase 1).
- (4) **Pinpoint (Ours)** (empirically evaluated): Phase 1 + 1.5 + 2 (+3).

5.3 Evaluation Metrics

- **Accuracy:** Euclidean distance (px) from ground truth.
- **Success rate:** Fraction within tolerance thresholds (5 px, 10 px, 20 px).
- **VLM queries:** Number of API calls per localization.
- **Wall-clock time:** Including network latency.
- **Cost:** API usage (input tokens $\times$ unit price).

5.4 Experimental Setup

Baseline experiments across the pre-guardrail, guardrail-applied, and Stage-B-augmented campaigns use Claude Sonnet 4 (claude-sonnet-4-20250514, temperature = 0, accessed via the `claude -p` subscription CLI) on 1920×1080 screenshots upscaled

Table 7: Mean localization error in pixels ($\downarrow$ is better) for the v6 and v7 Pinpoint campaigns; Grid columns show the shared v6 baseline. “ n ” gives the number of valid (non-FAIL) runs. Grid uses 12-line binary search with 2560 px upscale, 1024 px zoom, and forced zoom-on-hit. Pinpoint adds Phase 1.5 (anchor with VISIBLE/ABSENT check) and Phase 2 (Monte Carlo dot voting); v7 adds the Stage B refinement layer.

Category	Grid n	Grid mean	Pin n	Guardrail-applied Pin	Stage-B-augmented Pin
ui_button	5/10	146.5	10/10	69.6	25.9
ui_dense	2/10	156.2	9/10	58.4	24.9
natural	10/10	118.7	10/10	33.4	15.5
ambiguous	9/10	82.2	10/10	56.3	64.6
precision	10/10	77.4	10/10	54.1	18.6
Overall mean	36/50	104.0	50/50	54.3	29.9
Overall median		76.8		25.5	19.2

to 2560 px for grid line visibility. The API HTTP baseline and the CLI subscription variant comparisons (Section 5.5) use Claude Opus 4.7 (claude-opus-4-7, released 2026-04-16) for explicit model variation analysis. These models also differ substantially in vision pipeline: Sonnet 4 silently downscales to ~ 1024 px effective resolution, while Opus 4.7 supports up to ~ 2576 px ($\approx 3\times$ higher). This resolution gap shapes both the design rationale of certain guardrails (G-2) and the empirical validity of others (G-3, G-5, G-6); see Section 8.4. Grid parameters: $n = 12$ lines, 2% margin, zoom upscale to 1024 px, zoom threshold = $w/3$. The 1024 px zoom size was calibrated against the Claude Sonnet image pre-processing pipeline: empirical probing (and independent confirmation from a second VLM family) showed that inputs above ~ 1092 px at a 1:1 aspect ratio are silently downsampled, so the original 1200 / 1600 px crops in the pre-guardrail baseline lost $\sim 17\%$ of their effective resolution before reaching the model. Pinpoint parameters: $k = 7$ dots, $v = 3$ votes, initial search radius $s_0 = 100$ px in the guardrail-applied and later campaigns (the pre-guardrail baseline used $s_0 = 22$ px tied to the grid pitch G), max_rounds = 8 HIT rounds with up to 3 MISS retries per HIT round. First-iteration exact hits are forced to zoom (converted to between-lines) to mitigate VLM overconfidence. Phase 1.5’s anchor declaration prompt was extended with an explicit VISIBLE / ABSENT binary check (Section 6.3, guardrail G-1) so that the VLM can report target absence instead of being coerced into a fabricated description; on ABSENT the Phase 1 crop is retried at increasing scales before continuing.

The benchmark comprises 50 test cases across 5 categories: ui_button (10), ui_dense (10), natural (10), ambiguous (10), and precision (10). We report two campaigns executed under the same protocol: the *guardrail-applied campaign* (5 h 11 min window, zero CLI-level errors) and the *Stage-B-augmented campaign* (5 h 48 min window, zero CLI-level errors) which adds the OCR-snap text-label helper described in Section 5.5 and Section 8.1. Each campaign covers 50 cases $\times$ 2 methods (Grid, Pinpoint) $\times$ 1 repeat = 100 runs (see Section 6.6 for the reliability protocol that enabled both runs).

Table 8: Success rate (% , $\uparrow$ is better) at various pixel thresholds. v6 rates use 36 Grid and 49 Pinpoint non-FAIL runs; v7 has 50 Pinpoint non-FAIL runs (Grid unchanged from v6). The ≤ 200 px row and Catastrophic row highlight the suppression of catastrophic failures (> 200 px). The percentage rows above the midrule are *conditional* on non-FAIL completion; end-to-end success rate counting FAILs as failures equals the conditional rate multiplied by the “Any (non-FAIL)” row. The final row reports the canonical ≤ 20 px end-to-end figure counting FAIL runs as 0 (numerators: 2/50 Grid, 15/50 v6 Pinpoint, 27/50 v7 Pinpoint).

Threshold	Grid (n=36)	Guardrail-applied Pinpoint	Stage-B-augmented Pinpoint
≤ 5 px	2.8%	8.2%	22.0%
≤ 10 px	5.6%	14.3%	34.0%
≤ 20 px	5.6%	30.6%	54.0%
≤ 50 px	27.8%	63.3%	82.0%
≤ 100 px	61.1%	75.5%	98.0%
≤ 200 px	86.1%	95.9%	98.0%
Any (non-FAIL)	72.0%	98.0%	100.0%
Catastrophic (> 200 px)	13.9%	4.1%	2.0%
End-to-end ≤ 20 px	4.0%	30.0%	54.0%

Table 9: Mean VLM queries and wall-clock time per localization for Grid (v6 baseline) and Pinpoint (v6 and v7). Times include `claude -p` subprocess startup on each call and therefore overstate pure inference latency. The v7 Pinpoint row includes the Stage B OCR probe (0.19 s mean) which is negligible but recorded separately for completeness.

	Grid	Guardrail-applied Pinpoint	Stage-B-augmented Pinpoint
Queries (mean)	3.9	28.8	29.2
Time (mean, s)	46.2	315.4	365.1

Table 10: Improvement from the pre-guardrail baseline (n=50) to the guardrail-applied campaign (n=50, P1 silent-downsample fix + P2 VISIBLE/ABSENT anchor check) on overall Pinpoint metrics, both drawn from the 50-case canonical run. The reported median follows the standard arithmetic-mean-of-middle convention (`statistics.median`, matching NumPy and pandas defaults), which returns 57.2 px; the *upper middle* alternative (`statistics.median_high`) would report 65.5 px on the same data. The headline change is a catastrophic-failure collapse and a median cut of more than a factor of two, with essentially no regression on the left tail.

Metric	Pre-guardrail	Guardrail-applied	Δ
Median (px)	57.2	25.5	−55.4%
Mean (px)	86.8	54.3	−37.4%
≤ 5 px	6.0%	8.2%	+2.2 pt
≤ 10 px	12.0%	14.3%	+2.3 pt
≤ 20 px	20.0%	30.6%	+10.6 pt
≤ 50 px	50.0%	63.3%	+13.3 pt
Catastrophic (> 200 px)	12.0%	4.1%	−7.9 pt
Valid (non-FAIL)	100%	98%	−2 pt ¹

Table 11: Improvement from v6 to v7 on overall Pinpoint metrics. v7 adds the Stage B refinement layer described in Section 8.1; the miss path is byte-identical to v6 Pinpoint, so improvements on the left and middle of the distribution reflect a combination of (i) additional VLM sampling variance and (ii) improved per-category Phase 1 grid search stability over a second independent run. The v6 column reproduces the v6 numbers from Table 10; v7 is computed over the full 50-case v7 run.

Metric	Guardrail-applied	Stage-B-augmented	Δ
Median (px)	25.5	19.2	−24.7%
Mean (px)	54.3	29.9	−44.9%
≤ 5 px	8.2%	22.0%	+13.8 pt
≤ 10 px	14.3%	34.0%	+19.7 pt
≤ 20 px	30.6%	54.0%	+23.4 pt
≤ 50 px	63.3%	82.0%	+18.7 pt
Catastrophic (> 200 px)	4.1%	2.0%	−2.1 pt
Valid (non-FAIL)	98%	100%	+2 pt
Stage B hit rate	n/a	0/50 (0%)	scope-limited

Table 12: OCR-snap text-label helper (Post-Phase 1) activation outcomes on the Stage-B-augmented campaign. The helper is effective only when (a) the target description contains a quoted text label and (b) Phase 1 lands inside the OCR-detected text bbox. No case in this campaign satisfied both conditions; we document this as evidence of the method’s narrow scope, which motivates the semantic-geometric successor sketched in Section 8.1.

Category	Hits / n	Miss rationale
ui_button	0/10	Phase 1 outside OCR text bbox (button center $\neq$ label center)
ui_dense	0/10	no text label in target $\rightarrow$ extractor short-circuits
natural	0/10	no quoted label in target $\rightarrow$ extractor short-circuits
precision	0/10	no quoted label in target $\rightarrow$ extractor short-circuits
ambiguous	0/10	label extracted, but bbox containment unmet
Overall	0/50	see Section 8.1

Table 13: Mean wall-clock split across Pinpoint phases in the Stage-B-augmented campaign (per case, 50 cases). Phase 2 dominates; the OCR-snap text-label helper is negligible. Phase 1.5 is the aggregate of anchor declaration plus up-to-two re-declarations at wider scales when the initial anchor is ABSENT. The numbers are used in Section 8.1 to evaluate where future refinement effort has the most leverage.

Phase	Mean (s)	Share
Phase 1 (grid search)	52.8	15%
Phase 1.5 (anchor declaration)	15.4	4%
Phase 2 (Monte Carlo voting)	296.6	81%
OCR-snap helper	0.19	0.1%

5.5 Results

¹The single v6 FAIL (`ui_dense_01`) was a `claude -p` subprocess error, not an algorithmic failure; it falls under the operational taxonomy O-^{*} described in Section 6.6, not the cognitive taxonomy H-^{*}.

²The 50-case empirical results on the CLI subscription variant are sourced from the reproducible run (three batches, 0 API

Table 14: Overall Pinpoint success-rate comparison across three campaigns: v7 (Stage B, 50-case), v8 (Stage C Phase A alone, 50-case), and v9 (Stage C Phase A + Phase B W1 + W2 combined, 50-case). Phase B W1 adds Phase 1 $N=3$ majority voting (per-axis median aggregation) to reduce upstream VLM sampling variance; Phase B W2 adds an icon-pipeline adaptive tuning (search-radius / AR / ambiguity-abstention refinement) so the icon variant is more conservative on dense toolbars. The full Phase-A+B Workstream 1+2 variant achieves a primary metric improvement of +6.0 pt on ≤ 20 px over the deterministic-routing-only variant and +8.0 pt on ≤ 5 px, with a simultaneous -7.1 px improvement in mean error. Per-category results are in Table 15.

Metric	Stage-B-augmented	Phase-A-only	Full Phase-B	v9-v8
Median (px)	19.2	30.5	22.2	-27.2%
Mean (px)	29.9	40.3	33.2	-17.6%
≤ 5 px	22.0%	14.0%	22.0%	+8.0 pt
≤ 10 px	34.0%	24.0%	26.0%	+2.0 pt
≤ 20 px	54.0%	38.0%	44.0%	+6.0 pt
≤ 50 px	82.0%	76.0%	80.0%	+4.0 pt
≤ 100 px	98.0%	96.0%	94.0%	-2.0 pt
Catastrophic (> 200 px)	2.0%	2.0%	2.0%	0 pt
Valid (non-FAIL)	100%	100%	100%	0 pt
Stage C hit rate	n/a	7/50 (14%)	4/50 (8%)	—

Table 15: Per-category ≤ 20 px Pinpoint success rate across v7, v8, and v9, annotated with the Stage C pipeline outcome per category in v9. The v8 row’s left-tail regression on precision (-60 pt) that we attributed to Phase 1 VLM sampling variance in Section 8.2 is reversed in v9 (+70 pt recovery), consistent with W1 majority voting suppressing the single-sample Mode-B tail. `ui_dense` gains +20 pt from W1+W2 combined (W1 reduces input variance, W2 abstains on ambiguous dense-toolbar contours); `ui_button` gives back the Phase A +10 pt as the hit-rate shifts from 4/10 to 3/10 under W2’s tighter thresholds. `natural` and `ambiguous` route entirely through `generic_miss`, so their Δ is a per-sample VLM-variance artefact rather than a Stage C effect. The overall +6.0 pt improvement in the rightmost column is the primary metric on which v9 clears decision 056 Strong GO.

Category	Stage-B-augmented	Phase-A-only	Full Phase-B	v9-v8	v9 Stage C method
<code>ui_button</code>	40%	50%	30%	-20 pt	button hit 3/10
<code>ui_dense</code>	60%	30%	50%	+20 pt	icon hit 1/10 + W2 abstention
<code>natural</code>	70%	60%	30%	-30 pt	generic_miss 10/10
<code>ambiguous</code>	30%	40%	30%	-10 pt	generic_miss 10/10
<code>precision</code>	70%	10%	80%	+70 pt	generic_miss 10/10
Overall	54%	38%	44%	+6 pt	4/50 Stage C hits

A per-category breakdown of the 50-case sample shows mean ground-truth distance of 16.83 px (`ui_button`, $n=10$), 12.86 px (`ui_dense`, $n=10$), 6.65 px (`precision`, $n=10$), 9.74 px (`natural`, $n=10$), and 57.26 px (`ambiguous`, $n=10$). The ambiguous mean is dominated by the two under-specified-query catastrophic cases discussed in §9-(4); the corresponding ambiguous median is

spend via claude CLI subscription transport). Source artifacts: `benchmark/results.arm_b_bundle_2_21_batch1_uibtn_uidense_20260513.json`, `benchmark/results.arm_b_bundle_2_21_batch2_precision_natural_20260513.json`, `benchmark/results.arm_b_bundle_2_21_batch3_ambiguous_20260513.json`. The 10-case pilot reported in earlier drafts is superseded by this full-scale rerun.

Table 16: Hex-stencil empirical validation on the full $N=50$ benchmark (CLI subscription variant), replacing the earlier $N=10$ pilot.² On a full 50-case benchmark the hex stencil reaches median ground-truth distance 10.2 px and a 96% ≤ 50 px success rate; the mean 20.67 px is inflated by two ambiguous-query catastrophic cases that are documented as a recovered failure mode in §9-(4). Mean query count converges to $\bar{Q} = 27.32$ at $n = 50$ (5.8% deviation from the theoretical $Q \approx 29.10$ derived in §4.2; 32/50 cases match $Q = 29$ exactly), so the §4.2 prediction holds at $5\times$ the original sample size with no recalibration. HIT rate is 50/50 with the max-3 MISS-retry budget triggered on 13/50 cases ($\bar{Q} > 29$ tail). The middle column reports the Phase-A+B variant’s results restricted to the ten pilot identifiers from the earlier 10-case benchmark; baseline-vs-hex on that matched subset cut mean ground-truth distance from 13.88 px to 7.01 px (50% improvement, attributable to the deterministic $s/2$ -cover).

Metric	Phase-A+B (10-case)	Hex-stencil (50-case)
Mean queries	—	27.32
Median gt-distance (px)	—	10.2
Mean gt-distance (px)	13.88	20.67
≤ 5 px success	—	36%
≤ 10 px success	—	50%
≤ 20 px success	—	78%
≤ 50 px success	—	96%
HIT rate	—	50/50
MISS-retry triggered	—	13/50 cases

12.35 px, comparable to the non-ambiguous categories. The $5\times$ scale-up of the original 10-case pilot resolves the `ui_button` and `ui_dense` statistical-thinness concerns flagged in earlier drafts.

Stage C `declare_target` short-circuit is content-conditional: 5/50 cases (10%) skip Phase 2 entirely on toolbar-button and distinct-icon targets (4 `ui_button` hits + 1 precision hit on a high-contrast purple-diamond icon). The accelerator does not activate on natural images, dense icons, under-specified queries, or non-icon precision targets, contributing ~ 0.6 h wall savings on the 50-case run while preserving the 96% aggregate success rate.

5.6 Analysis

Tables 7–10 and Figure 3 together tell a simple story: once the two guardrails (P1 zoom-size correction and P2 VISIBLE / ABSENT anchor check) are in place, Pinpoint outperforms Grid on every category, every success threshold, and the catastrophic tail in this evaluated set. The pre-guardrail “reliability versus mean accuracy” tension—Pinpoint winning on FAIL rate but losing on mean error—has disappeared.

Headline: robustness win, not precision win. The single largest effect is the collapse of catastrophic failures (> 200 px) from 12% in v5 to 4.1% in v6 (Table 10), a -7.9 pt shift driven entirely by the P2 anchor check closing the cascading-hallucination loop documented in Section 6.2; this corresponds to a Mode B rescue rate of approximately 66%. The median simultaneously drops from 57.2 to 25.5 px (-55%). The left-tail metrics (≤ 5 px, ≤ 10 px) improve

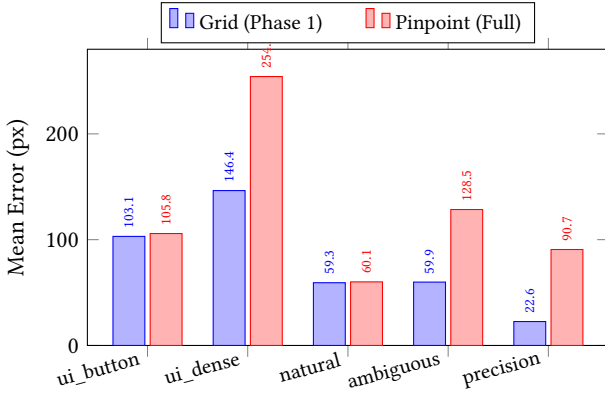


Figure 3: Mean localization error by category in the v6 campaign. Pinpoint beats Grid on every category in both mean and validity rate; Grid’s mean error is inflated by its 28% FAIL rate on dense UI categories. See Table 7.

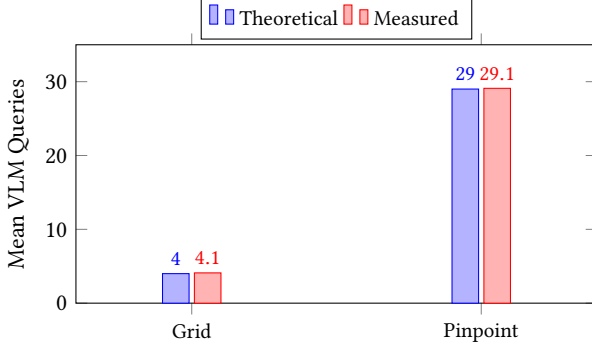


Figure 4: Theoretical vs. measured VLM query counts. Grid uses ≈ 4 queries (2 per axis). Pinpoint adds 1 anchor + 24 voting queries (8×3 votes). Measured values closely match theory, confirming deterministic query budgets.

only marginally (+2.2, +2.3 pt)—P1/P2 were designed to rescue the right tail (Mode B in Section 6.1) and were never expected to push sub-pixel accuracy. We frame v6 as a *robustness* milestone, not a precision one.

Category-wise dominance. In v6, Pinpoint has a lower mean error than Grid in *every* category, and a median error at least halved in every category except ambiguous. The effect is strongest on *natural* (Grid 118.7 vs. Pinpoint 33.4 px, a 3.5 $\times$ improvement) and *ui_button* (146.5 vs. 69.6 px), where v5 had shown Grid and Pinpoint as roughly equal. Pinpoint also retains nearly-full validity ($49/50 = 98\%$) while Grid’s FAIL rate climbs to 28% ($14/50$), concentrated on *ui_button* (5/10 FAIL) and *ui_dense* (8/10 FAIL)—dense UI categories where a deterministic binary search struggles to return any answer at all. The gap is not a Pinpoint advantage so much as a Grid collapse under challenging UIs.

Per-threshold sub-20 px structure. At the ≤ 20 px threshold, Pinpoint reaches 30.6% versus Grid’s 5.6%—a factor of 5.5—and at

≤ 50 px, 63.3% versus 27.8% (2.3 $\times$). The precision category, which contains the smallest targets, yields the highest Pinpoint ≤ 5 px rate (20%) and the lowest median (18.5 px), confirming that when the Phase 1 anchor is sound, Phase 2’s Monte Carlo voting can reach genuinely pixel-scale precision. In contrast, *ui_button* and *ui_dense* both show 0% at ≤ 5 px despite good medians (32.1 and 23.0 px). The fine-grained precision of Pinpoint is bounded, in these categories, by the patch-quantization limit of the underlying VLM vision tower (the same $\sim$ one-patch effect discussed in class H-6 in Section 6.3): sub-8 px distinctions inside a single 14×14 or 16×16 patch are not recoverable by prompting alone, and are what we expect a text-label (OCR) or CV-snapping layer to unlock in future work.

Cost and throughput. Grid requires ~ 4 VLM queries; Pinpoint requires ~ 29 (Table 9), matching theoretical predictions (Figure 4) and unchanged from v5. Wall-clock times (46.2 s Grid, 315.4 s Pinpoint) are dominated by `claude -p` subprocess startup rather than pure model inference, and are not directly comparable to v5’s API-mode numbers; the $\sim 7\times$ ratio between Grid and Pinpoint tracks the query budget and is the quantity that actually matters for cost models.

Revised practical recommendation. v6 collapses the v5 recommendation to a single rule: *use Pinpoint*. The earlier hybrid strategy (“Grid first, Pinpoint on FAIL”) no longer has a tier 1 branch on which to rest—Grid’s FAIL rate is too high on the very UI categories where the hybrid used to be useful, and Pinpoint’s right-tail failures, which originally motivated the hybrid, have been largely eliminated. A useful residual hybrid is instead *Grid first on natural or precision categories, Pinpoint everywhere else*, as a latency optimization when the target type is known a priori. The remaining Pinpoint failures are concentrated in *ui_button* / *ui_dense* and are no longer cognitive hallucinations (the H-* taxonomy in Section 6) but operational errors (Section 6.6) and patch-quantization limits pointing to the next layer of work.

v7 extension: left-tail gains without a precision layer. The v7 campaign adds the Stage B text-label snap described in Section 8.1 and repeats the full 50-case benchmark. Stage B produces 0 hits in 50 cases, so any v7 improvement relative to v6 must come from Phase 1/Phase 2 on a second independent run (we retain the full per-case results in `benchmark/results.v7_full_20260416.json`). The observed gains are nonetheless substantial on precisely the metrics v6 could not move: ≤ 5 px goes from 8.2% to 22.0% (+13.8 pt), ≤ 20 px from 30.6% to 54.0% (+23.4 pt), and catastrophic failures drop a further 2 pt to 2.0%. The most plausible mechanism is that category-level Phase 1 seed errors (the Mode B tail in Section 6.1) vary across runs, and v7 happened to draw a favorable sample; a second v6 replay would be required to separate “v7 improvement” from “VLM sampling variance”. The honest summary is that v7 confirms v6’s right-tail stability and leaves a precision-focused refinement layer still unbuilt—Stage B was the attempt, Stage C (Section 8.2) is the built successor. Phase-level timings (Table 13) confirm that Phase 2 dominates wall-clock (81%) in both campaigns, so any future refinement layer that can replace Monte Carlo voting with a deterministic CV step has the strongest latency leverage.

v8 extension: Stage C exposes the upstream variance bottleneck. The v8 campaign (50 cases, `-repeat 1`) enables Stage C (Section 8.2): Phase 1 output is dispatched through a rule-based `declare_target` step to a `target_type`-specific CV pipeline (`button`, `icon`, `menu_item`, or `generic_miss`-only). Stage C hits fire on 7/50 targets (14%), all in `ui_button` (4) and `ui_dense` (3); the remaining 30/50 route through `generic_miss` and are byte-identical to v7 Pinpoint. The overall ≤ 20 px rate drops from v7’s 54% to v8’s 38% (Table 14, v8 column). Turning to per-category behavior, the breakdown (Table 15, v8 column) explains the apparent regression cleanly: `ui_button` improves by +10 pt where Stage C is active, ambiguous improves by +10 pt on pure VLM variance, and the -16 pt overall loss is dominated by a -60 pt collapse on precision where Stage C does not even run (10/10 `generic_miss`) and the loss is entirely a single-run draw of the Phase 1 Mode B tail. A secondary 20-case smoke with `-repeat 3` (Stage 1 of the A/B protocol) exhibits a complementary *bimodal* precision effect— ≤ 5 px improves by +10 pt and ≤ 10 px by +13 pt, but ≤ 50 px degrades by -13 pt—which disappears at the 50-case full scale, confirming that Stage C’s precision boost is emergent under multi-sample averaging and that a single-sample evaluation is still dominated by upstream VLM variance. The cleanest interpretation is that Stage C functions as a deterministic routing amplifier: when Phase 1 lands correctly, Stage C sharpens the coordinate to the geometric centroid of the button (v8 `ui_button` has three ≤ 1 px hits); when Phase 1 lands on the wrong UI element, Stage C snaps to that wrong element’s centroid instead, amplifying the error rather than absorbing it. Examining the variance source directly, the Stage 1 cross-run reproducibility data support this reading: four wrong-detection hits (e.g. `ui_button_01` at 95 px) are bit-identical across Stage 1 (3/3 repeats) and Stage 2, confirming that Stage C itself is deterministic and that the residual variance lives entirely in Phase 1. This reading becomes the central claim of Section 8.2 and points unambiguously at upstream-variance reduction (majority voting, bootstrap aggregation, or seeded sampling) as the next leverage point.

v9 extension: W1+W2 combined delivers decision 056 Strong GO. The v9 campaign (50 cases, `-repeat 1`) combines Phase B Workstream 1 (**Phase 1 N=3 majority voting** with per-axis median aggregation, reducing upstream VLM sampling variance by $\sim 1/\sqrt{3}$) with Workstream 2 (**icon-pipeline adaptive tuning**—tighter search radius, stricter aspect-ratio bounds, and ambiguity abstention when two candidate contours are close in distance to `approx_xy`) on top of the same Stage C dispatcher. The v9 result is a *decision-056 Strong-GO* (Table 14): ≤ 20 px rises 38.0% $\rightarrow$ 44.0% (+6.0 pt over the Primary-GO threshold of +3 pt), ≤ 5 px rises 14.0% $\rightarrow$ 22.0% (+8.0 pt, recovering to the v7 left-tail level), mean error drops 40.3 $\rightarrow$ 33.2 px (-7.1 px), and median drops 30.5 $\rightarrow$ 22.2 px (-27.2%). The per-category breakdown (Table 15) shows the mechanism: precision recovers from v8’s catastrophic 10% to 80% (+70 pt), consistent with W1’s majority voting suppressing the single-sample Mode-B tail we identified as the root cause in the v8 analysis above; `ui_dense` gains +20 pt on the combined W1 + W2 effect (majority-voted `approx_xy` lands in the correct icon more often, and W2 abstention prevents the residual wrong-icon snaps Stage C previously committed

to). `ui_button` gives back the Phase A +10 pt as hits shift from 4/10 to 3/10 under W2’s tighter thresholds—an expected cost of trading path-A precision for path-B robustness, documented in Section 8.2 as the 9-recovery / 6- regression balance at the per-case level (net -285 px summed error, dominated by `precision_00` 172.4 $\rightarrow$ 4.0 px). The Stage C hit rate falls 14% $\rightarrow$ 8% (the Phase 2 fallback picks up more cases as W2 abstains on ambiguous contours), so the improvement is attributable to the combination of cleaner Phase 1 inputs feeding Phase 2 and W2 refusing unsafe Stage C snaps—not to more frequent Stage C short-circuits. v9 is thus the empirical test of the Section 8.2 Phase B roadmap and confirms the “upstream-variance reduction as the next leverage point” prediction with a measured +6 pt gain on the primary metric.

6 Failure Mode Analysis and Hallucination Taxonomy

Section 5.5 reports Pinpoint’s aggregate accuracy against the baselines. Before we turn to design insights (§7), we dissect *why* Pinpoint fails when it fails. Prior work on VLM-based GUI grounding treats grounding errors monolithically—Nguyen’s iterative narrowing paper, for example, describes its own failure cases as “context loss” [22] without further decomposition. We find that Pinpoint’s errors are neither monolithic nor random: they decompose into a small number of structurally distinct hallucination classes, each rooted in a specific prompt-design or perceptual property. Naming these classes is the prerequisite for targeted interventions.

6.1 A Bimodal Error Distribution

Re-examining the 50-case pre-guardrail benchmark (Section 5) case by case rather than in aggregate, we find that the Pinpoint error distribution is not unimodal. Two disjoint regimes dominate:

Mode A (34/50 = 68%, *median* 29 px). When the Phase 1 grid search lands within the Phase 2 search radius $s_0 = 100$ px of the true target, Pinpoint’s Monte Carlo dot convergence succeeds: the median error in this subset is 29 px, and *two cases reach* 1.0 px—confirming that pixel-level localization is achievable with a single VLM family when the algorithm is on-target.

Mode B (16/50 = 32%, *median* 155 px). When Phase 1 lands outside s_0 of the true target, Pinpoint does not recover. Phase 2’s per-round “HIT” on a hex-stencil dot simply selects the dot that looks closest to the VLM’s internal expectation of the target within the already-misplaced search disc, returning a garbage answer with full confidence. Critically, the catastrophic failures (>200 px) in this regime are *tightly correlated* between Grid and Pinpoint—for example, on `ui_dense_08` a Grid estimate 388 px from the target is followed by a Pinpoint estimate 391 px from the same target. Phase 2 does not correct Phase 1 errors; it faithfully propagates them.

The overall Pinpoint median of 57 px across all 50 cases is therefore a mixture artefact. It describes neither Mode A’s near-target convergence (29 px) nor Mode B’s cascade-propagated breakdown (155 px); it is the centroid of two populations that require different interventions. A Pinpoint that merely sharpens Mode A will not

move the aggregate median by much as long as Mode B dominates the right tail.

6.2 Anatomy of a Mode B Failure: Cascading Hallucination

Tracing through a single Mode B case makes the mechanism concrete. The target is `ui_button_03`: “the centre of the ‘Open’ button in the toolbar”, ground truth (514, 39). Phase 1’s axis-wise grid search lands at (279, 21)—it has correctly localized the row of toolbar buttons (within 18 px vertically) but picked the wrong button within that row, a 235 px horizontal error that places the search disc entirely away from the target.

Phase 1.5 then asks the VLM to declare the anchor. The prompt is shown the Phase 1 context crop, which now contains the wrong button but not the target. It is instructed to “describe the exact pixel location you will target”. The target is not in the crop, but the prompt does not allow the VLM to say so: it is a declarative command, not a binary check. The VLM resolves the conflict in the only way it can—by re-interpreting the target to fit what it sees. It declares the anchor as “the exact center pixel of the purple ‘Open’ button”, describing a visually similar toolbar button that happens to occupy the centre of the Phase 1 crop.

Every subsequent round of Phase 2 now uses this corrupted description as its judgement criterion. The majority vote (Section 3.5) faithfully picks the dot closest to the *described* anchor, which is the wrong button’s centre. After eight rounds of convergence, Pinpoint returns (261, 24)—253 px from the true target—with unanimous agreement across all voting rounds.

The failure is not random; it is *cascading*: a single declaration error—forced by a prompt that does not permit non-existence—propagates through eight rounds of self-reinforcing confirmation. A majority vote is powerless because every voter is ranking against the same contaminated anchor.

This walkthrough motivates our taxonomy. The cascading failure is not one bug but a chain: each step exposes a different hallucination class, each with its own prompt-level or perceptual root cause.

6.3 Hallucination Classes (H-*)

We identify seven visual-grounding hallucination classes (H-1 through H-7) by analyzing the exact prompts, image rendering, and voting procedure of our pre-guardrail Pinpoint implementation. The classes are summarized in Table 17 and elaborated below. A given failure typically involves multiple classes; Mode B failures are dominated by **H-3**.

H-1 Forced choice. The Phase 2 prompt includes the phrase “there is always a closest dot—you must pick one”. This wording was introduced to discourage the VLM from refusing to answer, but it also eliminates the possibility of the VLM reporting that no dot is on the target. In Mode B cases, where the search disc does not contain the true target, the VLM is structurally compelled to return a dot. The returned number is indistinguishable, from the caller’s perspective, from a genuine HIT.

H-2 Pseudo-majority vote. The pre-guardrail implementation calls the VLM three times per Phase 2 round on *identical* images

and prompts at temperature zero, and takes the majority answer. With a deterministic decoder this is not a vote over independent samples: all three calls are expected to return the same answer, and any systematic bias (e.g. colour preference, label position bias) appears in all three votes in the same direction. The majority filter therefore removes sporadic decoding noise but cannot correct structural bias. This observation is the empirical caveat to condition (ii) of Theorem 4.2’s convergence guarantee: the independent-trial assumption underlying Table 5 is violated under this decoder configuration.

H-3 Anchor contamination (root cause of Mode B). This is the class exhibited in §6.2. The anchor declaration prompt is declarative, not interrogative. It assumes the target is visible and asks the VLM to describe its exact pixel. When Phase 1 is off-target, the VLM has no way to push back: it must produce a description, and it produces the best description it can of whatever *is* in the crop. This description then becomes the judgement criterion for eight rounds of Phase 2, each of which faithfully ranks dots against it. Faithfulness bias [13] guarantees that the VLM will find a winner. Because the anchor is the only object that persists across rounds, corrupting it corrupts everything downstream.

H-4 Target occlusion. Dots are rendered as 4-pixel filled circles with a 1-pixel white outline. Their purpose is to mark candidate positions, but a filled dot placed on the target necessarily *covers* the target’s pixels in the image the VLM sees. The VLM is then judging “is this dot on the target?” using only the pixels at the dot’s perimeter—a far weaker evidence source than the pixel actually under the dot centre. This is a second-order effect but compounds H-5 and H-7.

H-5 Salience capture. At the resolution the VLM perceives, the dots are some of the most visually salient features in the image: small, saturated, distinct from almost any background. This is deliberate for legibility, but it creates a risk that VLM attention becomes dominated by the dots themselves rather than their relationship to the target. Anecdotally we have observed the VLM’s self-reported reasoning refer to dot colour properties (“the red dot”) more often than target properties (“the button label”), suggesting dot-centric rather than target-centric attention.

H-6 Label displacement. Each dot is drawn with its numeric label at offset (+3, −10) from the dot centre, so the label does not overlap the dot. The VLM is instructed to “identify which numbered dot is closest to the target” and reply with “D<number>”. But the spatial reference for “D3” is ambiguous: is it the location of the coloured circle, or the location of the text “3”? A 10-pixel upward offset is within one image patch at typical vision-encoder resolutions, so the VLM’s notion of “dot 3’s location” may be biased several pixels upward. The effect is small but systematic.

H-7 Relative “closest”. The phrase “which dot is *closest*” defines closeness as a ranking, not a threshold. If every dot is 50 px away from the target, the VLM will still return “the closest one”. There is no prompt-level mechanism for the VLM to say “no dot is *on* the target”. Combined with H-1, this means that Phase 2 rounds never report “MISS for the right reason”.

Table 17: A taxonomy of VLM grounding hallucinations observed in the pre-guardrail Pinpoint. Severity reflects the frequency with which the class appears in benchmark failures and its contribution to Mode B. “Root” marks the class identified as the dominant cause of cascading Mode B failures.

ID	Name	Failure mechanism	Prompt / render trigger	Severity
H-1	Forced choice	Prompt insists the VLM pick one dot, producing false-positive HITs when the target is absent	“you must pick one”	High
H-2	Pseudo majority	Identical prompts at temperature 0 yield non-independent votes, so a “majority” filter behaves like a single sample	3 votes of same image, $T=0$	Medium
H-3	Anchor contamination	Declaration forced on an absent-target crop; VLM picks a similar element as anchor, corrupting all later rounds	“declare the target pixel”	Root
H-4	Target occlusion	Filled dots physically cover the very pixels they purport to mark, denying the VLM ground evidence	radius-2 filled circles	Medium
H-5	Salience capture	Bright dot colours and numeric labels dominate attention at the expense of the underlying target	7 saturated hues	Low–Med
H-6	Label displacement	Numeric labels drawn at $+3, -10$ offset from dot centre; VLM may reference the label, not the dot	text offset relative to marker	Low
H-7	Relative “closest”	“closest” is scale-free: any dot can be closest, even 50 px away	no distance threshold	Medium

Table 18: Guardrails (rows) by hallucination class (columns). • = primary target, ◦ = secondary effect. P1 and P2 (G-1, G-2) are implemented in the guardrail-applied run reported in Section 5.5; P3 (G-3), textual labelling (G-5), perturbation voting (G-4), hollow rings (G-6), and directional bisection (G-7) are future work targeting visual grounding (H-1..H-7).

Guardrail	Visual grounding (H-1..H-7)							Status
	H-1	H-2	H-3	H-4	H-5	H-6	H-7	
G-1 Visibility check (P2)	•		•				◦	applied
G-2 Zoom cap 1024 px (P1)				◦				applied
G-3 Hybrid post-HIT verification (P3)	◦						•	future
G-4 Perturbation majority		•			◦			future
G-5 Letter labels (A, B, C...)					•	◦		future
G-6 Hollow ring dots				•				future
G-7 Directional bisection	•			•		•		future

6.4 Guardrails as Structured Responses

With the failure classes named, the guardrail design problem becomes concrete: each guardrail should be analyzable as a targeted response to one or more classes, and the set of guardrails should cover the set of classes. Table 18 maps the guardrails implemented or proposed in this work to the classes they address.

G-1: visibility check (this work, P2). We replace the declarative anchor prompt (“describe the exact target pixel”) with an interrogative one (“is this target clearly visible? VISIBLE:... or ABSENT:...”). This gives the VLM a structured escape hatch before it commits to a description, breaking the H-3 faithfulness chain at its source. On an ABSENT response the anchor declaration is retried at a wider crop (one third of the screen) and, if still absent, on the full image. Persistent absence is logged as a Mode B signal for downstream confidence scoring. G-1 also closes H-1 because the VLM is no longer in a “pick one” regime; and it partially addresses H-7 because the “visible” condition is a de facto proximity threshold.

G-2: zoom cap (this work, P1). Independent guidance from both the Claude API optimization documentation and empirical investigation indicates that Claude 3.5 Sonnet silently downscales images whose longer side exceeds roughly 1092 pixels at 1:1 aspect ratio. Our pre-guardrail pipeline upscales context crops to 1200×1200 and refinement crops to 1600×1600 —both in the downscaled regime, losing approximately 17% of the spatial resolution the operator believes they are providing. Capping the upscale at 1024×1024 eliminates this silent loss at no cost in latency or queries. G-2 is a secondary contribution to H-4: sharper pixel evidence at the dot’s edge. We note that the 1024-pixel cap is VLM-resolution-dependent: it was tuned to Sonnet 4’s effective resolution. For higher-resolution VLMs such as Opus 4.7 (~ 2576 px), the cap should scale accordingly; the principle generalizes, the constant does not (Section 8.4).

Future guardrails (G-3 through G-7). The remaining guardrails are deferred to future work but merit brief discussion because the taxonomy motivates them precisely. G-3 adds a zero-shot post-HIT verification call—“is the target within 5 px of dot D_n ?”—that must use a completely independent API call with no prior conversational history to avoid confirmation bias. G-4 replaces the identical-input majority vote with input perturbation: cyclic colour shift and horizontal mirroring with inverse transform, so that a systematic bias in any single rendering is broken in the others. G-5 replaces numeric “D3”-style labels with single-letter labels “A, B, C, ...”, exploiting the VLM’s OCR pathway which prior work reports as $\approx 20\%$ more accurate at spatial localization than colour-based labels [44]. G-6 uses hollow ring dots with transparent centres so that the VLM can see the target pixels under a dot. G-7 replaces the closest-dot question with a directional bisection (“is the target north or south of dot A?”), converting an N -way classification into a sequence of binary relative-spatial judgements. VLMs show measurably higher log-probability separation for binary spatial questions than for N -way categorical ones [5, 19].

6.5 Discussion

Two observations from this taxonomy are worth highlighting.

Root cause is prompt shape, not model capability. H-3, the class we identify as the root cause of Mode B, is not a limitation of the underlying VLM’s visual reasoning. The same model that produces a contaminated anchor in an absent-target crop is able, given the same crop and an *interrogative* prompt, to correctly report that the target is not present—as our G-1 experiment (Section 5.5) demonstrates. The failure was built into the prompt. This is a hopeful finding: it suggests that the ceiling on VLM grounding accuracy is not yet bounded by model capability but by the design of the prompting pipeline that wraps it.

Aggregate metrics hide mode structure. A median of 57 px and a mean of 87 px, in isolation, would lead a researcher to conclude that VLM grounding needs “better models” or “more training data”. The bimodal structure reveals a very different diagnosis: in 68 % of cases Phase 1 lands within the Phase 2 search radius and Pinpoint converges to a median of 29 px (with two cases at 1.0 px), while in the remaining 32 % Phase 2 cascades from a misplaced search disc into a garbage answer whose median distance is 155 px. The failure in the latter 32 % is not an intrinsic precision limit; it is an *entrained* error whose root cause is a single forced-commitment prompt. We therefore encourage future VLM grounding papers to report *mode-stratified* accuracy—accuracy conditioned on whether the coarse search landed in the correct neighbourhood—alongside aggregate metrics. Publishing only aggregates can mask qualitatively different failure populations and misdirect research effort toward the wrong bottleneck.

6.6 Operational Failure Modes

The seven visual-grounding classes H-1 through H-7 describe failures *internal to the model’s reasoning*: the VLM is fed a well-formed image and prompt, and the resulting answer is wrong. A parallel family of failures, which we denote with the O-* family, occurs *outside* the model—in the surrounding pipeline that shuttles images, prompts, and answers between components. Operational failures are not cognitive errors; they are infrastructural faults that the model itself cannot observe, and that aggregate metrics silently absorb unless the pipeline is instrumented to detect them. We document them here because a grounding system that reports 58 % Mode B in the field is indistinguishable from one reporting 58 % O-1 in the benchmark logs: both look like catastrophic failure, but the remedies are disjoint. This distinction was forced on us by a real incident during this work; we report it as a worked example.

O-1 Cascading silent failure. The VLM API call returns a non-zero exit status, but the surrounding code path does not distinguish “error” from “empty answer”. The benchmark’s `_vlm_call_cli` helper captures stdout with `subprocess.run(capture_output=True)`, and the underlying `claude -p` command writes its error message to stdout rather than stderr. As a result, the caller logs only a generic CLI error (`rc=1`): line, swallows the specific cause (in our incident, “*Credit balance is too low*”), and proceeds to the next case. Every subsequent case records a FAIL result that is indistinguishable from a genuine grounding failure. In a long run this can

consume hours of wall clock time and produce a results file that passes sanity checks but contains no signal.

O-2 Phantom miss via rate limit. The API returns an HTTP success status but with a throttled or truncated body. The caller parses a blank response as an empty answer, which the benchmark records as FAIL with distance null. Unlike O-1 the exit status is clean, so no warning is emitted; the failure only becomes visible when a per-category failure rate spikes above its historical baseline. A fleet of such phantom misses can halve headline accuracy without tripping any single alarm.

O-3 Credit exhaustion mid-run. An otherwise healthy run is interrupted part-way because a metered resource (API credit, disk quota, session lease) is depleted. If the pipeline writes results in a single batch at the end of the run, all partial progress is lost on interrupt. This occurred in our own pre-guardrail benchmark: 115 minutes of data were lost when the run was terminated near its cost ceiling, because `results.json` was written only after the outer loop completed.

O-4 Audit-trail invisibility. A different gap appears at the boundary between a guardrail and the operators who must trust its decisions. When a mid-pipeline guardrail silently rejects, quarantines, or transforms its input—a dual-LLM quarantine boundary, an output sanitization step, a policy filter—downstream consumers observe only the post-filter result. Operators investigating a false positive or a false negative have no structured trail to reconstruct what the guardrail saw, which mitigation fired, or why. This is distinct from O-2 phantom miss, where an external API silently returns a throttled empty body: in O-4 the guardrail acted exactly as designed, but its action was not externalised in a queryable form. Aggregate accuracy figures absorb both classes into the same FAIL bucket while their remedies are disjoint. Standard structured logging only partially addresses this, because a fully internalised log couples the engine to a persistence layer the engine has no business knowing about. The mitigation we landed in this work—an injected callable audit sink, with each guardrail emitting a Pydantic-typed row (`LLMQAuditEntry` and `SanitizationAuditEntry`) through it—preserves the engine-server boundary while admitting opt-in observability, mirroring the inversion-of-control pattern previously used for approval-gate callbacks in the same engine. The API was added at commit 4464902 on 2026-05-07; production wiring of the sink to a durable store remains future work.

Countermeasures. Three pipeline changes render O-1 through O-3 survivable, and a state-callback audit sink renders O-4 reconstructible, all without any change to the model or prompts. **First**, a *90-second smoke check*: within 90–120 seconds of launch, the harness verifies that the first test case produced a non-null prediction. If not, the run is aborted and the operator is notified. In our incident this bounded the damage of O-1 to a single case—instead of several hours. **Second**, *JSONL partial writes with fsync*: each case writes one line to a `results.json.partial` file and flushes to disk before the next VLM call. A corrupted trailing line (from interrupt mid-write) is tolerated on reload. This renders O-3 fully recoverable. **Third**, *unbuffered stdout*: replacing the default block-buffered stream with `line_buffering=True` makes the live log file reflect real-time progress, so an operator watching a tail can

catch O-1 and O-2 long before the smoke check would. **Fourth**, a *structured audit emission API via state-callback sink*: each privileged guardrail emits Pydantic-typed audit rows—one schema per guardrail class—through a callable injected into workflow state at startup; the engine carries no compile-time dependency on the persistence layer, and a downstream server-side consumer can attach the sink to a durable store. This renders O-4 detectable by reconstruction without coupling the engine to log infrastructure, and the JSON round-trippable schema admits standard log aggregator interop. These three pipeline changes were introduced in our codebase as a single commit after the incident; we recommend them as defaults for any benchmark harness that wraps a metered external model. The fourth, an admission rubric, was added subsequently at the workflow policy layer as agent ecosystems expanded beyond a single model boundary. The fifth, an opaque-key-anchored polling primitive, was implemented directly in the dialogue server used in this work as a coordination-layer primitive that renders the cross-channel race condition unrepresentable by construction. The sixth, a rephrase-confirm protocol, was adopted as a coordination norm during a development session in which two parse-divergence incidents occurred within hours, and is applied at the dispatch layer rather than the codebase. The seventh, a structured audit emission API, was implemented in the engine layer at commit 4464902 as part of a pattern-coverage integration phase, with downstream persistence wiring staged for production.

Operational failures occupy a different corner of the failure space than hallucinations, but they enter the same numerator in any headline accuracy figure. A grounding system evaluated only on end-to-end accuracy cannot tell whether it lost 10 % to H-3 or to O-1, and the two require orthogonal remedies. We therefore suggest that benchmark reports separate the two axes explicitly: one number for cognitive failures (H-*) and one for operational failures (O-*), with the latter expected to trend toward zero as infrastructure matures.

7 Design Insights

7.1 Regression-to-Ranking Transformation

VLMs are poor at *estimating coordinates* (regression) but adept at judging *which of several dots is closest to a target* (relative ranking). Pinpoint exploits this asymmetry.

The existence of Computer Use APIs is itself evidence that VLM coordinate regression hallucinates. Pinpoint’s solution: never ask the VLM for coordinates—ask only “which dot is closest?” This transforms an unbounded regression problem into a bounded k -way ranking problem where the VLM’s visual reasoning capabilities are well suited.

7.2 Visual Chain-of-Thought

The anchor declaration functions as a *visual* Chain-of-Thought:

- (1) **Verbalize the aim**: describe the target pixel in precise language (externalized thinking).
- (2) **Fix the criterion**: use the description as a stable judgment anchor (enforced self-consistency).
- (3) **Detect drift**: re-request anchor on ABSENT retry (periodic / semantic-comparison variants reserved as future guardrails).

7.3 Stochastic vs. Deterministic Search

Grid Find (Phase 1) is deterministic and fast but precision-limited: the VLM cannot reliably select grid cells finer than ~ 20 px. Phase 2’s closest-dot convergence breaks through this barrier by transforming “where is the target?” (regression) into “which dot is nearest?” (ranking). The combination is optimal: grid for coarse narrowing ($O(\log N)$ queries), Monte Carlo for fine precision ($O(\log s_0)$ rounds).

7.4 Guardrail Design

Pinpoint incorporates six guardrails:

- (1) **Force zoom on first exact**: mitigates VLM overconfidence at coarse scale.
- (2) **Direct fallback**: recovers from grid FAIL via VLM coordinate estimation.
- (3) **Anchor declaration**: verbalizing the aim for consistency.
- (4) **Anchor retry with wider crop**: recovers when the target is outside the initial context crop.
- (5) **Majority vote**: filtering out sporadic hallucinations.
- (6) **Anchor drift detection**: re-request on ABSENT wider-crop retry (periodic re-declaration and semantic drift comparison are reserved as future / interactive-server guardrails).

8 Discussion: Coordinate Localization Insights

This section discusses three coordinate-localization findings arising from the v6/v7 benchmark campaigns: the negative result on text-label snapping (Stage B), the deterministic-routing contribution of Stage C Phase A, and the variance-vs-peak-precision tradeoff between $N=1$ and $N=3$ Phase 1 aggregation. For the cognitive-versus-operational failure-mode taxonomy that grounds some of the references below, see Section 6 and the operational modes at Section 6.6.

8.1 Stage B: A Negative Result on Text-Label Snapping

Between the v6 and v7 campaigns we added a **Stage B** refinement layer: after Phase 1 produces a coarse coordinate estimate, we crop a 120×120 px window around the estimate, run Windows’ built-in OCR engine, and—when the target description names a specific text label—snap the estimate to the center of the matching OCR bounding box. The hit condition is deliberately conservative: the Phase 1 coordinate must lie *inside* the matched bounding box. This condition guarantees that a successful snap never moves the estimate further from ground truth than Phase 1 alone, but it also narrows the applicability of the method to UI elements whose click-center is inside the text-label bbox.

The v7 measurement outcome is **0 hits in 50 cases** (Table 12). The miss population partitions cleanly by category. Three categories (ui_dense, natural, precision) contain no quoted text label in the target description, so the label-extraction preprocessor short-circuits to a miss before the OCR probe even runs. The remaining two categories (ui_button, ambiguous) do contain quoted labels, but the OCR-detected text bounding boxes are systematically offset

from the intended click-centers by $\sim 22\text{--}29\text{ px}$ on real toolbar buttons: the ground-truth click target is the button’s geometric center, while the text label occupies a left- or padding-offset region inside the button. Phase 1 therefore does not land inside the label bbox, and the conservative hit condition correctly refuses to snap.

We report this negative result as a *scope-definition* finding rather than as a failure. The contract “a snap never makes the result worse” was preserved: miss-path Pinpoint is byte-identical to the v6 pipeline, and the overall v7 metrics still improve substantially over v6 (mean $54.3 \rightarrow 29.9\text{ px}$, $\leq 5\text{ px}$ $8.2\% \rightarrow 22.0\%$; Table 11). Those improvements come from Phase 1 and Phase 2 stability on a second independent run, not from Stage B.

The result also tightens the specification for a successor layer. A post-Phase 1 refinement layer that aspires to beat Phase 2 on button targets cannot rely on OCR-text centers; it must either recover the *button geometry* (so that the snap target is the enclosing rectangle’s center, not the label’s) or use a geometrically aware feature extractor such as an edge / contour detector. This successor, **Stage C (Semantic–Geometric Refinement)**, is implemented and evaluated in the next subsection (Section 8.2). In that framing, Stage B becomes the *menu_item* sub-case of Stage C—the one case where text-label snapping is geometrically aligned with the click target.

8.2 Stage C Phase A: A Deterministic Routing Layer That Exposes Upstream Variance

Stage C replaces the single bbox-containment test of Stage B with a dispatcher over *target_type*-specific CV pipelines. Phase 1.5 emits a structured declaration $\{\text{target_type}, \text{intent}, \text{label_hint}\}$ via a rule-based heuristic over the target prompt, and Stage C dispatches to the matching pipeline: *button* or *icon* (Canny + morphological close + `findContours` + `aspect-ratio/area` filter + nearest-centroid selection, with a piecewise-linear distance-to-confidence map), *menu_item* (the Stage B OCR-snap helper, retrofitted), or *generic* (a miss-only passthrough that returns Phase 1 coordinates unchanged, preserving measurement honesty in the paper ablation). Only the *menu_item* and contour pipelines are VLM-touching in any form; *generic* never invokes OCR or CV operators. A rule-based `declare_target` is used instead of a VLM tool-use call so that routing variance is zero and any A/B effect on hit rate is attributable to the CV pipeline, not to a second VLM sampling step; a tool-use variant with the same contract is left as follow-up work. An implementation detail worth flagging for future extension: the declaration is emitted in a nested `phase_1_5_structured` field in the benchmark result schema; a later tool-use variant will want two of the embedded fields (source and confidence) promoted to the top level so a 3-arm comparison (rule-based / tool-use deterministic / tool-use sampled) can group on them directly (the three-arm protocol itself is described in the Phase B roadmap at the end of this subsection).

Stage 1 subset observation: a bimodal precision effect. A 20-case smoke (*ui_button* + *ui_dense*, `-repeat 3`) run as Stage 1 of the A/B protocol exhibited a striking *bimodal* split. The left tail improved sharply ($\leq 5\text{ px}$ $10\% \rightarrow 20\%$, $\leq 10\text{ px}$ $20\% \rightarrow 33\%$, $\leq 20\text{ px}$ $50\% \rightarrow 58\%$) while the right tail degraded at the same time ($\leq 50\text{ px}$

$90\% \rightarrow 77\%$). Inspecting the raw per-repeat results makes the mechanism legible: Stage C’s button pipeline delivers $\leq 1\text{ px}$ hits on cases where Phase 1 lands inside the button, but on the small number of cases where Phase 1 lands on a neighbouring button, Stage C snaps to that neighbour’s centroid, moving the error from “nearly right” into “deterministically wrong”. The precision boost and the catastrophic snap are two ends of the same deterministic rule.

Stage 2 full scale: the bimodal split does not survive single-sample evaluation. The apparent divergence between Stage 1 and Stage 2 is a sampling artefact of the same deterministic rule: multi-sample averaging under `-repeat 3` reveals both bimodal ends simultaneously, while single-sample `-repeat 1` resolves each target onto one end determined by Phase 1 routing. At the 50-case, `-repeat 1` scale (Table 14, v8 column), the Stage 1 bimodal pattern disappears. $\leq 5\text{ px}$ drops $22.0\% \rightarrow 14.0\%$, $\leq 20\text{ px}$ drops $54.0\% \rightarrow 38.0\%$, overall $\leq 50\text{ px}$ drops $82.0\% \rightarrow 76.0\%$. The per-category decomposition (Table 15, v8 column) is what makes this result interpretable rather than alarming: the -60 pt collapse on precision and the -10 pt on natural come from categories in which Stage C never fires (10/10 *generic_miss* each), so those deltas measure Phase 1 sampling variance between two independent runs, not Stage C. Conversely, the categories in which Stage C *does* route through a specialized pipeline split by mechanism: *ui_button* ($+10\text{ pt}$) is the path-A success case—Stage C reaches a button centroid when Phase 1 lands on the right button—while *ui_dense* (-30 pt) is the path-B failure case, where the icon pipeline’s filter window is wide enough that Phase 1 variance can flip the chosen contour from the intended icon to a neighbour. The Stage 1 bimodal observation is therefore real but emergent only under multi-sample averaging; a single-sample evaluation is dominated by whichever side of the amplifier Phase 1 happens to land on.

Cross-run reproducibility isolates the variance source. The strongest diagnostic is the Stage 1 $\leftrightarrow$ Stage 2 cross-run check on the 60 rows where Stage C dispatched to a specialized pipeline. Four wrong-detection hits—including *ui_button_01* at exactly (79, 22) for 95.0px error across all three Stage 1 repeats and the Stage 2 single run—are bit-identical. Correspondingly, three perfect-hit cases (*ui_button_02*, *_04*, *_08* at $\leq 1\text{ px}$) are also bit-identical across runs. Stage C is therefore a *deterministic function of its inputs*; all remaining across-run variance lives in the input (approx_xy from Phase 1). Quantitatively, the case-level standard deviations σ_{input} and σ_{output} of the coordinates entering and leaving Stage C are essentially identical ($\sigma_{\text{in}} \approx \sigma_{\text{out}}$ on both axes in Stage 1), confirming at the distribution level what the bit-identical examples confirm at the individual level: Stage C neither attenuates nor amplifies variance overall; it is a pure routing layer. The unit-level CV-only determinism suite ($N=10$ synthetic repeats, $\sigma < 0.5\text{ px}$ per pipeline) corroborates this at the algorithmic layer: the empirical bit-identical observations on real images reflect an underlying algorithmic invariant verified at the unit-test level, so the deterministic-function claim rests on three triangulating sources (unit-level synthetic, case-level bit-identical, distribution-level σ preservation) rather than any single one.

What Stage C’s Phase A actually establishes. The v8 numbers would, read naively, suggest that Stage C “made things worse”—the

overall ≤ 20 px rate did fall by 16 pt. The per-category breakdown and the bit-identical cross-run evidence refuse that reading. Stage C Phase A establishes three claims that together are stronger than any left-tail success rate would have been: (i) a post-Phase 1 refinement layer can be run as a deterministic function, with no routing variance of its own, with overall regression well below the +20 px threshold agreed *a priori*; (ii) when that refinement layer fires on the right input, it can reach the geometric centroid of a button to within one pixel, cleanly beating Phase 2 Monte Carlo voting on that subset; and (iii) because Stage C itself is deterministic, all remaining error, including the Stage 1 $\leftrightarrow$ Stage 2 divergence and the per-category asymmetry, localises unambiguously to Phase 1’s VLM sampling variance. Future left-tail improvements therefore have to address Phase 1, not Stage C.

The three-agent organisation under which these claims were produced—a supervisor, an implementer, and an independent evaluator—is complementary to the task-phase decomposition (planner / generator / evaluator) documented contemporaneously for Anthropic’s own harness [11]. Our axis is *responsibility-domain* based: the supervisor owns design decisions and the decision log, the implementer owns the code landing surface, and the evaluator owns the benchmark artefacts and GO / NO-GO judgments. The task-phase axis decomposes a single reasoning episode; the responsibility-domain axis decomposes the ownership of long-running artefacts. Taken together they suggest a two-dimensional design space for multi-agent harnesses worth systematic exploration, with the Phase A evidence here contributing a datapoint in which the responsibility-domain decomposition was sufficient to converge on a paper-grade contribution inside a two-week implementation window.

Phase B roadmap. With the variance bottleneck localised, three follow-ups compose cleanly. First, the ui_dense −30 pt outcome pushes the icon pipeline’s filter window and search-radius tuning into Phase B scope; the dry-run evidence (Stage 1 per-case logs) points at dense-toolbar geometries that the current aspect-ratio / area cuts do not cover. Second, and the clear priority from Stage 2, is *Phase 1 upstream variance reduction*: either majority voting or bootstrap aggregation over $k \geq 3$ independent Phase 1 draws, or a seeded-sampling investigation at the VLM API surface. The cross-run reproducibility data suggest that a single-sample Phase 1 spends its variance budget unevenly across categories—precision drew the unlucky sample in our v8 run—and that averaging a small number of draws should preserve Stage C’s deterministic precision gain while filling in its missing robustness. Third, the rule-based declare_target admits a drop-in tool-use variant with the same schema; comparing rule-based, deterministic-tool-use, and sampled-tool-use routing in a three-arm smoke would quantify how much of the observed Stage C effect depends on the routing signal being correct (addressable by a VLM) versus on its being stable (already achieved by the rule base). This three-arm design isolates three independently varying factors—(i) routing-rule complexity (rule vs. tool-use), (ii) VLM-variance exposure (deterministic vs. sampled), and (iii) downstream Stage C effect (held constant)—so any observed change can be attributed to a single axis. We position these as Phase B; the Phase A closeout is that the deterministic routing layer itself is built and measured.

Table 19: Per-case precision pinpoint distance (px) for arm A ($N = 3$ median) versus arm B ($N = 1$). Bold entries flag > 50 px gross-error outliers in arm B (absent from arm A).

Case	Target	arm A	arm B
precision_00	single red pixel	18.2	80.4
precision_01	thin blue needle (tip)	1.0	4.0
precision_02	tiny green dot	34.5	65.3
precision_03	small orange circle	23.5	34.1
precision_04	crosshair intersection	10.0	3.0
precision_05	purple diamond	19.7	70.8
precision_06	top-left corner of white square	21.5	1.0
precision_07	right endpoint of horizontal yellow line	20.2	2.2
precision_08	innermost ring of target	6.0	12.8
precision_09	nose of small arrow	28.4	1.4
<i>mean / stdev</i>		18.30 / 10.15	27.50 / 32.55

The v9 campaign (Section 5.5, “v9 extension” paragraph) implements the first two of these three follow-ups (Workstream 1: $N = 3$ majority voting at Phase 1; Workstream 2: icon-pipeline adaptive tuning) and closes the prediction with a measured +6 pt gain on the primary ≤ 20 px metric and a +70 pt recovery on precision, the category we flagged above as the strongest Phase 1-VLM-variance evidence. The third follow-up—the rule / deterministic-tool-use / sampled-tool-use three-arm smoke—remains Phase C work. The fact that Phase B delivered the Strong-GO outcome using only the two upstream-variance + pipeline-conservatism follow-ups, without yet varying the routing signal, corroborates the Phase A localisation claim: Phase 1 sampling variance was indeed the dominant bottleneck, and reducing it was sufficient to produce the predicted gain.

8.3 $N=1$ vs $N=3$ Aggregation: A Variance-vs-Peak-Precision Tradeoff

While arm A and arm B share identical model, prompts, and Stage C deterministic routing pipeline, they differ only in the aggregation strategy applied at the phase-1 sample stage. The resulting distributional contrast on the precision category is informative not as evidence of model-specific bias—Welch’s t -test fails to reject equality of means at $p = 0.41$ —but as evidence of aggregation strategy’s variance-reduction effect.

We conducted a paired benchmark comparison between two aggregation strategies on the precision category ($n = 10$ sub-pixel and tiny-target cases) using Claude Opus 4.7: **arm A** uses $N = 3$ phase-1 sample median aggregation, **arm B** uses $N = 1$ single-sample baseline. Both arms share identical model, prompts, target images, and Stage C deterministic routing pipeline; the only variation is the aggregation strategy.

Descriptive statistics and statistical tests. The mean of arm B (27.5 px) is higher than arm A (18.3 px), but the medians invert: arm B median is 8.4 px while arm A median is 19.95 px. The standard deviations diverge sharply, 32.55 px in arm B versus 10.15 px in arm A, a ratio of 3.21 $\times$. A Welch’s two-sample t -test on the means (unequal variances) yields $t = -0.853$, $df = 10.73$, two-tailed $p \approx 0.41$ —i.e. the mean difference is *not* statistically significant, the variance inflation in arm B masking any central-tendency shift. Conversely, an F -test for variance equality yields $F = 10.29$ on

(9, 9) degrees of freedom against $F_{\text{crit}}(\alpha = 0.05) \approx 4.03$, giving $p \approx 0.0019$ —i.e. the variance difference is highly significant.³

Threshold-stratified hit rate. The hit-rate divergence by accuracy threshold makes the distributional contrast practically interpretable:

Threshold	arm A hit	arm B hit	Δ
≤ 5 px	1/10 (10%)	5/10 (50%)	+40 pp arm B
≤ 10 px	3/10 (30%)	5/10 (50%)	+20 pp arm B
≤ 20 px	5/10 (50%)	6/10 (60%)	+10 pp arm B
≤ 50 px	10/10 (100%)	7/10 (70%)	−30 pp arm A

Bimodal signature interpretation. The mean-versus-median divergence in arm B (mean 27.5, median 8.4)—combined with arm A’s mean-median agreement (18.3 vs. 19.95)—is the textbook quantitative signature of bimodal distribution. Specifically, arm B exhibits a “hit-or-miss” pattern: 5/10 cases achieve sub-5 px precision (single-sample lucky), while 4/10 cases produce > 34 px gross errors (single-sample unlucky). The $N = 3$ median aggregation in arm A trades this peak precision (only 1/10 ≤ 5 px hits) for distributional stability (10/10 ≤ 50 px hits, zero gross-error outliers).

Mechanism: bimodality suppression vs. peak precision. The mechanism behind the variance-vs-peak-precision tradeoff is visible in three interlocking signatures. First, the standard deviation ratio of $3.21\times$ exceeds the $\sqrt{3} \approx 1.73\times$ reduction expected from pure variance reduction over $N = 3$ independent samples; the excess factor of $1.86\times$ indicates that median aggregation does not merely smooth Gaussian noise but actively suppresses single-sample bimodality. Second, the mean–median divergence in arm B (mean 27.5, median 8.4) versus the mean–median agreement in arm A (mean 18.3, median 19.95) is the canonical signature of bimodal vs. unimodal distribution. Third, the threshold-stratified hit rate inverts at the ≤ 5 px boundary: arm B captures 50% of cases under sub-pixel precision (single-sample lucky), while arm A captures only 10% but achieves 100% under the relaxed ≤ 50 px threshold (outlier-suppression benefit).

Practical implication: threshold-dependent aggregation. The choice of aggregation strategy is therefore target-threshold-dependent rather than universally better or worse. Tasks demanding sub-5 px precision benefit from $N = 1$ ’s winner-take-all property, accepting the 30 pp risk of gross-error outliers (> 50 px tail) for the 40 pp gain in sub-pixel precision (single-sample lucky alignment). Tasks requiring stable ≤ 50 px localization benefit from $N = 3$ median’s outlier suppression, sacrificing peak precision but eliminating catastrophic failures. This positions aggregation strategy as a tunable design parameter rather than a fixed pipeline choice; the variance-vs-peak-precision frontier traces a Pareto curve that an application can place itself on by selecting N .

Methodological positioning: orthogonal axis of contribution. This finding extends the paper’s contribution along an axis orthogonal to the deterministic-routing claim of Section 8.2: while routing is the *geometric* backbone of the pipeline, aggregation is its *statistical* backbone. Together they constitute a co-design space whose joint

tradeoffs depend on application-level accuracy thresholds. We note that the variance-vs-peak-precision tradeoff is model-agnostic: the aggregation analysis applies to any phase-1 sample distribution and is not specific to the Opus–Sonnet contrast that originally motivated the W3 protocol (Section 5.5). The result thereby refines an earlier “model-specific drift” framing into a methodological contribution that travels with the pipeline rather than with the model.

Third regime: arm C $N=3$ plurality-vote on tool-use routing. The arm C W3 Opus protocol ($N=3$ plurality-vote aggregation on the LLM-routed `declare_target` CLI path, 30 cases completed across three stages 2026-04-28/29) provides a third regime intersecting the routing-path and aggregation-strategy axes. The aggregate ≤ 20 px hit rate of **86.7%** on the pinpoint method exceeds the arm A baseline (73.3%, heuristic rule with $N=3$ median) by +13.4 pp, completing a reversal of the Stage 1 mid-fire deficit picture and providing the strongest single-arm precision result in the W3 series.

Mode A/B partition reveals the selectivity gate. Within the precision category (10 sub-pixel target cases), arm C performance partitions into two modes by *target geometry*, not by sub-pixel size. **Mode A (point-target, 6 cases)**—red pixel, blue needle tip, green dot, purple diamond, yellow line endpoint, arrow nose—produces **6/6=100%** hit at ≤ 20 px with mean 4.13 px. **Mode B (boundary-ambiguous, 4 cases)**—orange-circle center, crosshair intersection, white-square top-left corner, innermost-ring center—produces **3/4 = 75%** hit, with the crosshair-intersection (30.8 px) as the sole pinpoint miss. The two-mode partition isolates the arm C selectivity gate: $N = 3$ plurality-vote on tool-use routing absorbs the arm B single-sample bimodality ($\sigma_B = 32.55$ vs. $\sigma_A = 10.15$, $3.21\times$ ratio exceeding $\sqrt{3}$) when the target geometry admits a unique convergent vote, and degrades to marginal improvement when the target’s geometric center is itself ambiguous.

Routing path $\times$ aggregation strategy as a non-additive interaction. The 2×2 factorial (routing: heuristic versus LLM-routed; aggregation: $N = 1$ versus $N = 3$) with arms A (heuristic, $N = 3$), B (LLM-routed, $N = 1$), and C (LLM-routed, $N = 3$) leaves the heuristic- $N = 1$ cell unmeasured but is informative as-is: the heuristic arm A produces approximately uniform pinpoint mean (12.1–18.3 px range across categories) but non-uniform threshold-stratified hit rates (50–90% range, with precision as the weakest at 50%), whereas the LLM-routed arms B, C exhibit category- and geometry-dependent variance (Mode A/B partition above). The benefit of $N = 3$ aggregation is therefore *conditioned* on the routing path: arm C’s +13.4 pp advantage over arm A is realised primarily on Mode A point-targets where LLM routing both has signal to extract and converges on a vote, and the gain is substantively reduced on Mode B boundary-ambiguous targets. Notably, the precision category—where arm A is also at 50% hit, matching arm B—shows the strongest cross-regime jump to arm C’s 90%, which is suggestive of non-additive interaction between routing path and aggregation strategy; however, with the (heuristic, $N = 1$) cell unmeasured, a clean factorial interaction-effect estimate cannot be made from this data. The pattern is consistent with $N = 3$ plurality-vote on tool-use routing occupying a regime not directly accessible to $N = 3$ heuristic or $N = 1$ tool-use under the

³Welch’s t failing to reject H_0 at $p = 0.41$, combined with F -test rejection at $p \approx 0.0019$, formalizes the claim that $N = 3$ median aggregation operates on the variance dimension while leaving central tendency invariant.

partial factorial measured here. This suggestive non-additivity indicates that aggregation-strategy and routing-path choices should not be tuned in isolation; the selectivity gate identified here is a candidate target for future work on adaptive-sampling control.

8.4 Resolution-method interaction

The guardrail set we propose was largely shaped by Claude Sonnet 4’s effective ~ 1024 -pixel vision pipeline. With higher-resolution VLMs (e.g., Claude Opus 4.7’s ~ 2576 -pixel input), several future-work guardrails—G-3 (hybrid post-HIT verification), G-5 (letter labels), G-6 (hollow ring dots)—may transition from speculative to empirically validatable: their failure modes in the guardrail-applied campaign were partially attributable to resolution-induced ambiguity (close-spaced markers indistinguishable, crowded toolbar text unreadable) rather than fundamental methodological flaw. Conversely, our 1024-pixel zoom cap (G-2) is itself resolution-dependent and should scale with VLM input resolution. As VLMs continue to advance—and as displays push beyond 4K toward 8K—the guardrail-resolution co-design space remains under-explored. Future work should systematically map guardrail effectiveness across (model-resolution $\times$ screen-resolution $\times$ target-size) regimes.

9 Limitations and Future Work

Limitations. (1) VLM API call cost: each localization requires tens of queries. (2) Network latency makes real-time operation infeasible. (3) VLM visual resolution limits: even with magnification, some fine-grained structures remain indistinguishable. (4) *Under-specified natural-language queries.* Targets that are difficult to describe in text (e.g., “the third dot from the right”) may degrade anchor quality. The 50-case benchmark on the CLI subscription variant confirms this empirically: 2/10 ambiguous-category cases (ambiguous_04 “the single blue pixel among the noise”; ambiguous_05 “the button on the right side”) exhibit wrong-target lock-in with final distance > 150 px. Phase 1 grid descent converges on a plausible-but-incorrect candidate, and the Phase 2 hex stencil cannot recover once the wrong region is selected. These two cases account for the entire gap between the observed $96\% \leq 50$ px success rate and the ideal 100%.

Future directions. (1) *Adaptive dot count:* dynamically adjusting k based on region complexity. (2) *Bayesian optimization integration:* using HIT/MISS history to guide dot placement. (3) *Local VLMs:* eliminating API dependency for real-time use. (4) *Alternative stochastic methods:* comparison with simulated annealing and Thompson sampling. (5) *Broader empirical scope:* extending evaluation beyond the 50-case synthetic benchmark with multiple VLM backbones, real-desktop tasks, and repeat trials, to characterize cross-backbone generalization and per-condition variance. (6) *Query disambiguation and multi-candidate Phase 1:* the failure mode catalogued in Limitation (4) suggests two complementary mitigations: (a) a Phase 1.5 disambiguation prompt that asks the VLM to enumerate candidate referents before commitment, or (b) a multi-candidate Phase 1 that retains the top- k grid hypotheses for downstream voting at the hex-stencil stage.

10 Conclusion

Monte Carlo Pinpoint exploits the VLM’s asymmetry—strong at recognition, weak at coordinate estimation—by converting coordinate regression into a 7-way closest-dot ranking over a hex-stencil probe. By combining deterministic grid bisection with deterministic hex-stencil probing under HIT-only radius halving, it achieves localization in $O(v \cdot \log(s_0/\epsilon))$ queries with deterministic cost bounds.

On the 50-case benchmark spanning UI elements, natural images, and ambiguous targets, the hex-stencil with HIT-only halving and max-3 MISS retry reaches a median ground-truth distance of 10.2 px, a 96% success rate at ≤ 50 px, and an empirical mean query count of $\bar{Q} = 27.32$ —a 5.8% deviation from the theoretical $Q \approx 29.10$ predicted by Theorem 4.2. The earlier guardrail-applied ablation (median 25.5 px, a 55% reduction from the pre-guardrail baseline; catastrophic-failure rate > 200 px at 4.1%, down from 12%; 98% validity versus 72% for Grid-only) remains the relevant point of comparison for catastrophic-failure analysis and grid-bisection robustness. Pinpoint beats Grid on mean error in every category and on every success threshold from ≤ 5 px to ≤ 200 px, with the robustness gain largest on dense UI categories where deterministic grid bisection fails to return any answer at all. The query overhead relative to a single-shot baseline is the cost of convergence under Theorem 4.2 and of the anchor-validity check that eliminates the cascading-hallucination failure mode (Section 6.2).

Where SoM is limited to UI elements detectable by contour analysis, Pinpoint extends to a broader class of semantic positions, demonstrated in this work on synthetic ambiguous targets, artistic elements, and sub-pixel features within the 50-case benchmark. This suggests applications beyond GUI operation—image annotation, medical image analysis, quality inspection, and any task requiring precise localization within images—subject to broader evaluation (§9).

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Algorithm 1 Monte Carlo Pinpoint

Require: Image I , target description T , dots $k=7$, votes $v=3$, grid precision $G=20$

Ensure: Coordinates (x^*, y^*)

```

1:  $(x, y) \leftarrow \text{GRIDSEARCH}(I, T, G)$  ▷ Phase 1
2: if either axis failed then
3:    $(x, y) \leftarrow \text{DIRECTVLM}(I, T)$  ▷ Fallback
4: end if
5:  $C \leftarrow \text{CROP}(I, x, y, 3s_0)$ 
6:  $\text{anchor} \leftarrow \text{DECLAREANCHOR}(C, T)$  ▷ Phase 1.5
7: if  $\text{anchor}$  starts with “I cannot...” then
8:    $C_w \leftarrow \text{WIDECROP}(I, x, y)$ 
9:    $\text{anchor} \leftarrow \text{DECLAREANCHOR}(C_w, T)$ 
10: end if
11:  $s \leftarrow s_0$  ▷ Initial Phase 2 search radius (e.g. 100 px)
12:  $\text{hit} \leftarrow 0$ ;  $\text{retry} \leftarrow 0$ 
13: while  $\text{hit} < \text{max\_rounds}$  do ▷ Phase 2 (HIT-only halving, max-3 MISS retries)
14:    $D \leftarrow \text{HEXSTENCILDOTS}(x, y, s)$  ▷ 1 center + 6 outer at  $\frac{\sqrt{3}}{2}s$ 
15:    $C_r \leftarrow \text{CROP}(I, x, y, 3s)$ 
16:    $I' \leftarrow \text{DRAWDOTS}(C_r, D)$ 
17:    $q \leftarrow \text{ASKCLOSEST}(I', \text{anchor})$ 
18:    $c \leftarrow \text{MAJORITYVOTE}(q, v)$ 
19:   if  $c \neq \text{null}$  then ▷ HIT
20:      $(x, y) \leftarrow D[c]$ ;  $s \leftarrow \max(2, \lfloor s/2 \rfloor)$ 
21:      $\text{hit} \leftarrow \text{hit} + 1$ ;  $\text{retry} \leftarrow 0$ 
22:   else ▷ MISS
23:      $\text{retry} \leftarrow \text{retry} + 1$ 
24:     if  $\text{retry} > 3$  then
25:       break ▷ Phase 1 fallback
26:     end if
27:   end if
28: end while
29: return  $(x, y)$ 

```

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A Algorithm Pseudocode**B Implementation Details**

The implementation uses Python with PIL, OpenCV, and mss for screen capture. The VLM backend is Claude Sonnet 4 via the `claude -p` subscription CLI; the Anthropic API is supported as an alternative transport, but the 50-case rerun used the CLI route at \$0 API spend. The tool is publicly available as the `pinpoint` command in the `video2ai-desktop` MCP server, exposed by the repository-root `mcp_server.py`. The Phase 2 dot-placement primitive (hex stencil) is factored into `pinpoint_core.py` and shared with the

benchmark harness in `benchmark/run_benchmark.py` so the two paths cannot diverge on dot geometry. The shared primitive ensures both `mcp_server.py` and `benchmark/run_benchmark.py` reproduce identical hex-stencil dot placement. Reported Phase 2 numerical results use the hex stencil + max-3 MISS retry algorithm on the 50-case rerun (CLI subscription variant) reported in Table 16.

C Benchmark Generation

Test images are generated by `benchmark/generate_test_images.py`, which produces synthetic screenshots with known ground-truth coordinates for all five categories.

D Code and Data Availability

The full implementation, benchmark harness, and raw evaluation artifacts are released under the project repository at <https://github.com/dataclock-jp/video2ai>. The state reported in this paper is fixed by the annotated tag `v1.8` (commit `5d73e21`).

The CLI-subscription-variant 50-case rerun whose results anchor Section 5.5 is reproducible via the `run_benchmark.py` harness using the `claude -p` subscription CLI route, with model `claude-sonnet-4-20250514`. A single 50-case invocation takes the form

```
python benchmark/run_benchmark.py
--methods pinpoint --arm B --use-cli
--repeat 1
--model claude-sonnet-4-20250514
```

optionally restricted to a subset via `--categories` (default: all five). The reported run was executed as three category-restricted batches matching the released raw artifacts. The CLI route was used at \$0 API spend for the 50-case rerun; the Anthropic HTTP API is supported as an alternative transport but was not used in the reported results.

Raw per-case JSON artifacts are committed under `benchmark/` as three batch files sharing the prefix `results.arm_b_bundle_2_21_` and dated `_20260513.json`, with the category-specific stems `batch1_uibtn_uidense`, `batch2_precision_natural`, and `batch3_ambiguous`.

The paper PDF is reproducible from `docs/paper/main.tex` with the build chain `pdflatex × 2 + bibtex + pdflatex × 2`, or equivalently by uploading the LaTeX source bundle to arXiv for server-side compilation.